SHARD: cell-keyed residual splitting for alignment-resistant private dense retrieval

Sergey M. Kurilenko
Moscow Institute of Physics and Technology
kurilenko.sm@phystech.edu

June 6, 2026

Abstract

Dense embeddings underpin semantic search and Retrieval-Augmented Generation, yet a leaked vector store reveals much of the underlying text. The dominant modern attacks—few-shot alignment, zero-shot inversion and unsupervised cross-space translation—all exploit the same weakness: the protected store is a *single global geometry* that can be aligned to a known space. A secret global rotation, the usual lightweight defence, falls to orthogonal Procrustes once the attacker has on the order of the subspace dimension in known plaintext pairs. We measure this baseline and then introduce a transform built to remove its weak axis.

SHARD is a family of retrieval-preserving embedding transforms. The centred embedding is rotated and split into a short *public* prefix (used for coarse stage-1 retrieval) and a *private* residual that is sharded into C coarse cells, each rotated by its own secret orthogonal key; the residual is reranked under CKKS, where the per-cell keys cancel and preserve the exact inner product. A single parameter C interpolates between the global-linear baseline ($C{=}1$) and per-document micro-keys ($C{=}N$). On cached embeddings from five encoders we find: **(i) utility**—because reranking is full-dimensional, SHARD recovers the raw quality that the truncating baseline loses, including the nDCG@10 that half-SVD truncation drops on BEIR (e5-base/ SciFact: 0.638 vs. raw 0.637 vs. baseline 0.585); **(ii) alignment resistance**—recovering the private residual into the native frame under a *diffuse* known-plaintext leak requires $\approx C\times$ more anchors (m_{50} rises from 200 for a global key to 102,400 for $C{=}256$, reproduced on a second encoder), at a measured cost of $\approx$7–30 encrypted residual queries per search; **(iii) public-index leakage**—the short prefix reveals far less neighbour structure than the baseline's half-space (NN-overlap@10 0.20–0.55 vs. 0.76), and the per-document micro-key limit drives the residual graph leak to zero with an unlinkable, renewable template. The barrier is not a Procrustes artefact: learned-linear (ALGEN-core), non-linear and unsupervised (VEC2VEC-core) attackers do no better, and at matched utility a distortion-aware DP-noise defence leaves de-anonymisation R@1 near 1.0 where SHARD gives 0.0. We are equally explicit about the limits: cell-level keys leave within-cell similarities and a $\approx C\times$ query-cost trade-off; a *targeted* attacker who concentrates anchors in a victim's (public) cell needs only $\approx d_{\text{priv}}$ anchors; and an *overlapping* reference corpus still de-anonymises through the public prefix. SHARD hardens the diffuse alignment/inversion surface, not the coarse graph; it is an attack-aware geometric defence, not a cryptographic guarantee.

Keywords: dense retrieval, embedding privacy, protective transforms, alignment attacks, embedding inversion, cell-keyed splitting, cancellable templates, homomorphic encryption, CKKS, retrieval-augmented generation.

1 Introduction

The privacy of vector representations of text—"embeddings"—has become a first-class concern in production retrieval systems. Dense retrievers (DPR [5], ColBERT [6]) and state-of-the-art encoders (Sentence-BERT [1], multilingual-E5 [2], GTR [3], BGE-M3 [4]) push the per-document representation through a $\sim 10^2$-dimensional bottleneck that nevertheless preserves enough information to reconstruct the original text with high BLEU. Vec2Text [7] reports BLEU $\approx 97.3\%$ on 32-token inputs when the attacker has query access to the encoder; GEIA [8] and TEIA [9] show that even snapshot attacks (an adversary with read access to the vector database) recover personally identifiable information at high rates. For Retrieval-Augmented Generation pipelines whose vector stores typically contain customer support tickets, internal corporate documents and personal data, this rule of thumb—"a leak of the embedding is a leak of the text"—is now well documented [48, 18].

The defensive landscape is bracketed by two extremes that, individually, do not satisfy industrial requirements. Differential privacy [22, 23] adds calibrated noise per-record but, in the naive coordinate-Gaussian formulation on retrieval-quality embeddings, it faces a difficult operating-point choice: mild noise preserves much of the nearest-neighbour geometry, while stronger noise destroys ranking quality long before the privacy budget reaches a level considered useful. Fully homomorphic encryption [27, 33] allows arbitrary computation on ciphertexts, but the ciphertext-ciphertext (ct-ct) regime is too expensive for top-k search across 10^6 documents.

The weak axis of global-linear defences. Between the two extremes sits a popular lightweight middle ground: compress the collection onto a data-dependent SVD subspace, apply a single secret orthogonal rotation, publish an approximate-nearest-neighbour (ANN) index, and rerank the encrypted query under CKKS. We first measure this global-linear stack and confirm that its protection lives on *one weak axis*—the protected store is a single globally-aligned geometry. Concretely (Section 8): a known-plaintext orthogonal Procrustes attack recovers the secret rotation with about $k = d/2$ anchors, after which an overlapping reference corpus yields near-exact paragraph recovery (99.8% top-1); the public product-quantisation index preserves a mean cosine of 0.95 and 67% of exact top-10 neighbours; and because the rerank happens in the truncated half, the stack *loses* retrieval quality on real IR, dropping nDCG@10 by 2–8 points on BEIR. These are exactly the weaknesses that modern few-shot alignment [10], zero-shot inversion [11] and unsupervised cross-space translation [17] attacks exploit: a single global map is, in the end, alignable.

Shard: an attack-aware geometric transform. We introduce Shard, a family of retrieval-preserving protective transforms designed against the alignment and index-leakage attack surface rather than against a distortion surrogate such as σ_{rec} (which, as we and others show, is not a privacy metric). The construction (Section 7) keeps the CKKS/ct-pt machinery of the baseline but replaces its single global geometry. The contributions are:

- *Construction.* We rotate the centred embedding and split it into a short *public* prefix u (top-d_{pub} PCA, for coarse stage-1 retrieval) and a *private* residual r. The residual is sharded into C coarse cells defined by u, each rotated by its own secret orthogonal key H_c (a product of Householder reflections), giving the stored shard $z_i = H_{c(i)} r_i$. Reranking is done under CKKS in the residual space, where the orthogonal keys cancel ($\langle H_c r_q, z_i \rangle = \langle r_q, r_i \rangle$) and preserve the *exact* inner product. A single knob C interpolates between the global-linear baseline (C=1) and per-document micro-keys (C=N), and gives revocability at cell granularity.

- *Utility: full-dimensional rerank recovers raw quality* (Section 8.11). Because reranking is full-dimensional, Shard recovers the raw ranking that the truncating baseline loses. On the BEIR tasks where half-SVD truncation significantly drops nDCG@10, Shard matches raw (e5-base/SciFact 0.638 vs. raw 0.637 vs. baseline 0.585) while exposing a public channel only a quarter to a half the baseline's width.

- *Alignment resistance scales with C* (Sections 8.12–8.13)—the headline result, with its cost and caveats measured, not assumed. Under a *diffuse* known-plaintext leak, recovering the private residual into the native frame grows $\approx C\times$: median anchors rise from 200 (global key) to

25,600 (C=64) and 102,400 (C=256), and the result reproduces on a second encoder. The cost is $\approx$ 7–30 encrypted residual queries per search (the short-list spans that many cells); and a *targeted* attacker who concentrates anchors in a victim's public cell needs only $\approx d_{\text{priv}}$ (320/576 on our two encoders, vs. 1.7× the baseline's $d/2$), so the $C\times$ gain is an aggregate/diffuse-leak property. The barrier survives stronger attackers (Section 8.14)—the learned-linear core of ALGEN, a non-linear MLP, and unsupervised VEC2VEC-style matching all do no better than the orthogonal Procrustes we use—and against a distortion-aware DP-noise baseline at matched utility (Section 8.17), SHARD alone reaches near-zero de-anonymisation.

- *The public channel leaks coarse, not fine, structure, and micro-keys remove the residual leak* (Sections 8.15, 8.16). The short prefix reveals far less neighbour structure than the baseline's exposed half (NN-overlap@10 from 0.20 at d_{pub}=$d/16$ to 0.55 at $d/4$, vs. 0.76). The within-cell residual graph the server can reconstruct shrinks with C (0.25 at C=64, 0.18 at C=256) and reaches 0.00 for per-document micro-keys, which also give an unlinkable, renewable template (re-keying AUC $\approx$ 0.5).
- *Honest limitation* (Section 8.18). SHARD hardens the diffuse native-frame mapping, not the coarse graph: within a cell the keys cancel, and an *overlapping* reference corpus still de-anonymises through the public prefix once its single cheap global key is recovered. We measure this and discuss the coarse-cell-ID and micro-key variants that close it at a recall or bandwidth cost.

The global-linear baseline characterisation (alignment, PQ leakage, BEIR-truncation loss, and the calibrated-noise diagnostic that σ_{rec} is not a privacy metric) is retained in Section 8 as the foil that motivates SHARD, together with the CKKS ct-pt speed-up and parameter selection (Sections 8.1, 5) that both schemes share.

Notation. To avoid the overloading of the symbol N, we use N_{poly} for the CKKS polynomial-modulus degree, N_{docs} for the corpus size, and K_{cands} for the stage-1 short-list length throughout.

An elementary proxy bound. We state a one-line projection bound (Proposition 1): any decoder whose image is constrained to span(V_k) has L_2 reconstruction error at least $\|\mathbf{x}_\perp\|_2$. This is a Pythagorean fact, not a security theorem; we use it only to make σ_{rec} a precise distortion proxy within that decoder class. Its empirical relation to the BLEU of an off-the-shelf VEC2TEXT attack is reported as Empirical Hypothesis 1.

Roadmap. Section 2 reviews related literature; Section 3 states the threat model; Section 4 describes the global-linear baseline (the foil) and the shared CKKS/ct-pt machinery; Section 5 the CKKS parameter selection; Section 6 an elementary projection bound; Section 7 introduces SHARD; Section 8 reports the baseline characterisation and the four SHARD experiments (utility, alignment resistance, public-index leakage, and the reference-lookup limitation); Section 9 discusses limitations and open work.

2 Related Work

Embedding inversion. VEC2TEXT [7] trains an encoder–decoder model that iteratively rewrites a candidate hypothesis until its embedding matches a target vector; on 32-token MS-MARCO inputs the attack achieves BLEU $\approx$ 97.3% and $\approx$ 92% exact recovery. GEIA [8] relaxes the discreteness of the input by operating in the continuous embedding space and back-transcribing. TEIA [9] extends the threat model to training-data extraction. Earlier work of Song and Raghunathan [19] already established that embedding models leak substantial information about their inputs. Adjacent work on membership-inference [20] and on extracting training texts from large language models [21] confirms that the risk surface of dense retrievers is broader than the embedding-inversion literature alone suggests. The 2025 generation of attacks further weakens any defence that relies only on an unknown embedding orientation: ALGEN aligns a victim embedding space to an attack space with few known pairs and then generates text [10]; ZSInvert removes the need for an embedding-specific inversion model by using a universal zero-shot decoding strategy [11]; and LAGO extends few-shot inversion to cross-lingual settings by coupling alignments across related languages [12]. These papers motivate the alignment experiment in Section 8.7: if a secret rotation can be absorbed by a small alignment matrix, it must not be described as a security guarantee.

Lightweight defences and transformed representations. Recent defences increasingly treat embeddings as representations whose geometry can be compressed, transformed or regularised rather than merely noised. EGuard uses a projection network with mutual-information regularisation to reduce inversion leakage [13]; SPARSE studies concept-aware privacy mechanisms and anisotropic noise for embedding inversion resistance [14]; and nonparametric variational DP work adds bottleneck/noise mechanisms at the representation level [15]. Matryoshka-style representation learning shows that retrieval utility can be deliberately concentrated in prefix subspaces [16], while recent cross-space translation work argues that embedding spaces share enough geometry for transfer maps to be useful [17]. Shard builds on both observations: it uses a Matryoshka-style prefix as a deliberately coarse public channel, precisely because the cross-space-translation literature shows that a single global geometry is alignable.

Cancellable templates and decoupled encoders. Shard's cell-local keys import two ideas that are standard in biometric template protection but rare in text retrieval. Cancellable biometrics [32] require a protected template to be *renewable* (re-key after compromise) and *unlinkable* (the same identity under two keys should not be matchable); a product of secret Householder reflections per cell gives Shard both at cell granularity. Separately, work on retrieval with asymmetric document/query encoders [5, 6] shows that the document and query sides need not share one geometry—the structural premise that lets Shard key the stored side while keeping queries encrypted. Our contribution is to combine a coarse public prefix, a cell-keyed private residual, and CKKS reranking into one retrieval-preserving transform, and to show *measured* advantages on the alignment and index-leakage attack surface relative to the global-linear baseline.

Differential privacy for embeddings. Lyu et al. [24] investigate per-coordinate Gaussian noise on dense representations; the empirical finding is that retrieval quality degrades sharply long before ε reaches a useful range. Random-projection-based privacy [26, 25] predates modern dense retrieval; the data-dependent SVD projection used here is closer to a noise-removal technique than to a privacy-preserving DP mechanism, although σ_{rec} provides a useful proxy quantity.

Homomorphic encryption. The CKKS scheme [27] introduced approximate-arithmetic FHE; OpenFHE [33] and TenSEAL provide production-grade implementations. The bootstrapping cost [28] pushes practitioners to leveled circuits; specialised compilers [34, 35] optimise scale and modulus chains. GPU acceleration of CKKS bootstrap [36] and Intel HEXL [37] have made larger leveled circuits feasible.

Hybrid privacy-preserving retrieval. Tiptoe [38] combines clustering with PIR for query privacy. SealPIR [39], OnionPIR [41] and SimplePIR [40] provide single-server PIR with practical query sizes. The architecture in this paper is complementary: ct-pt CKKS reranking takes care of the value side of the leakage, while access patterns remain a separate problem for which PIR/ORAM-like primitives are an appropriate composition.

3 The Measured Setting (Threat Model)

Because the leakage measurements only make sense against a stated adversary, we fix the setting we measure: an asymmetric *trusted client / honest-but-curious server / non-adaptive attacker / unknown rotation* configuration. We then measure two ways in which this configuration can fail (known-plaintext alignment and PQ-code leakage), so the threat tables below double as a map of what our experiments do and do not cover. This is a fat-client, data-owner setting: the client runs the encoder, holds all keys, downloads the PQ index, and performs stage-1 retrieval locally; the server only reranks. That work/trust split is itself a scoping assumption and not the only possible deployment.

- The *client* is the data owner. It runs the encoder E, generates the secret keys $\mu, V_k, R, sk_{\mathrm{CKKS}}$ and never shares them. The client is trusted in software and key management.
- The *server* runs the search service over the rotated database $E_{\mathrm{rot}} \in \mathbb{R}^{N_{\mathrm{docs}} \times k}$ and the public PQ artefact. It is honest-but-curious: it follows the protocol but attempts to extract textual information from the data it sees and from the queries it processes.
- The *baseline attacker* is non-adaptive: it uses an off-the-shelf VEC2TEXT model trained without prior knowledge of R. We additionally evaluate a known-plaintext alignment attacker and PQ-code leakage, and then combine alignment with an off-the-shelf generative Vec2Text inversion test. Adaptive text decoders trained end-to-end against the rotated space and malicious server behaviour remain out of scope.

Three distinct privacy notions. A recurring source of confusion in hybrid designs is the conflation of *what* is protected. We therefore separate three independent notions and state, for each, exactly which component is responsible:

- **Document privacy.** The plaintext rotated database E_{rot} *is* visible to the server. It is protected only by SVD truncation (σ_{rec}, a proxy quantity) and the secret rotation R (an empirical obfuscation layer). *This is not a cryptographic protection*; it fails under known-R and, as Section 8.7 shows, under a modest known-plaintext alignment attack.
- **Query privacy.** CKKS hides the numerical value of the query vector and of the similarity scores from the server. This part *is* cryptographic, at the chosen security level.
- **Access-pattern privacy.** The candidate IDs reranked per query, and the public PQ codes, are *not* hidden. CKKS does not address this channel; it requires composition with a PIR/ORAM-style primitive.

In particular, the public PQ artefact (codebook + per-document codes, trained in the rotated space) is itself a lossy compressed view of E_{rot}. Section 8.7 quantifies this leakage on a controlled sample; the result is strong enough that public PQ codes must be treated as a meaningful side channel rather than as a harmless index artefact.

Table 1 summarises the coverage and Table 2 maps each claim to the evidence that supports it. The diagonal of the coverage table is intentional: ct-pt reranking takes care of the value channel, but the access-pattern channel (which document IDs are reranked) remains open, and the public-PQ-artefact channel is measured as leaky rather than closed. They would require composition with a PIR-style primitive and a stronger static-side index protection, respectively.

4 Method

The asymmetric construction under study is illustrated in Figure 1. The architecture is the union of an offline data-preparation phase (Section 4.1) and the online query protocol (Section 4.2).

Table 1: Threat coverage of the proposed protocol.

Attack vector	Covered?	Comment
Snapshot of E_{rot} without R (off-the-shelf Vec2Text)	Partial	Heuristic; empirically validated against non-adaptive Vec2Text.
Network interception, leakage of similarity scores	Yes	CKKS at tc128; sk never leaves the client.
Known R (known-rotation attacker)	No	Reduces to σ_{rec} from SVD truncation only.
Known-plaintext pairs $(\mathrm{text}_i, E_{\mathrm{rot},i})$	No	Evaluated in Section 8.7; roughly k pairs recover the rotation numerically.
Reference-corpus overlap after known-plaintext alignment	Partial	Evaluated in Section 8.8; overlapping texts are recoverable.
Unsupervised cross-space translation attacks	No	Not evaluated; recent embedding-space transfer attacks motivate separate stress testing.
Adaptive decoder retrained against rotated space	No	Sections 8.2 and 8.9 use off-the-shelf Vec2Text, not a learned decoder retrained on the protected space.
Malicious server (protocol deviation, substitution)	No	Verifiable computation / TEEs are an orthogonal addition.
Access-pattern leakage on candidate IDs	No	CKKS hides values, not access patterns; requires PIR/ORAM.
Leakage from the public PQ artefact (rotated-space codes)	Partial	Evaluated in Section 8.7; neighbour structure is substantially preserved, so this remains an exposed channel.

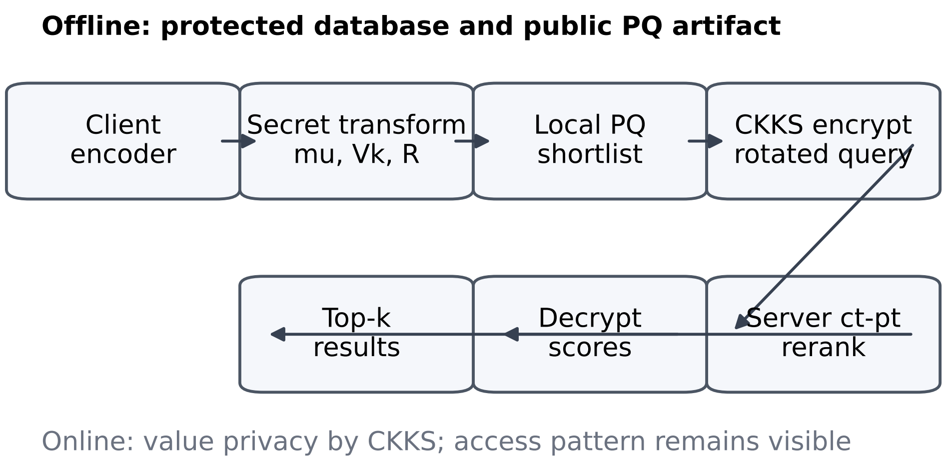

Figure 1: Online query flow of the construction under study. Steps 1–7 realise the two-stage protocol: client transformation $T(\cdot)$ and local PQ filtering produce a short-list of K_{cands} candidate IDs; step 4 encrypts the rotated query under CKKS; step 5 runs ct-pt reranking on the server; steps 6–7 decrypt the scores and sort. The secret keys (μ, V_k, R, sk_{CKKS}) and the rotated database E_{rot} are produced offline by the data owner (Section 4.1).

Table 2: Claims vs. evidence. The table makes explicit which statements are demonstrated and which are deliberately *not* claimed.

Claim	**Evidence / status**
CKKS hides query values and scores from the server	Cryptographic construction + parameters (Sec. 5, 4.3); query privacy is cryptographic.
ct-pt is faster than ct-ct for the reranking	Section 8.1 (1.44× on the micro-benchmark, 1.7× at the auto-tuned configuration).
Secret rotation helps against *off-the-shelf*, non-adaptive Vec2Text	Section 8.2; effect holds only for the tested weak-attacker configuration.
Aligned off-the-shelf Vec2Text recovers text/PII after Procrustes	Section 8.9: **not observed** in the memory-capped RTX 4090 run, but the raw baseline is also near the reconstruction floor, so this is not a proof of security.
Secret rotation is robust to known-plaintext alignment	**False**; Section 8.7 shows near-exact recovery with about k known pairs.
Public PQ codes are harmless metadata	**False**; Section 8.7 shows substantial reconstruction and nearest-neighbour leakage.
The protective wrapper preserves ranking in span(V_k)	Section 8.4 (within the 5-seed CI).
SVD truncation improves *Acc@1* (a "denoiser")	**Not significant**; Section 8.5: paired tests give $p \geq 0.11$ for every encoder.
SVD truncation significantly improves *Acc@10* for d=1024 retrieval-trained encoders	Section 8.5 (e5-large +0.042, $p < 10^{-6}$); it significantly *degrades* compact/paraphrase encoders.
The denoiser generalises to real IR	**False**; Section 8.6: SVD significantly *reduces* nDCG@10 on BEIR SciFact/NFCorpus for every encoder.
σ_{rec} alone measures document privacy	**Not claimed**; Section 8.3 shows that matched-distortion noise can preserve substantially more raw neighbour structure.
End-to-end query latency < 1 s at 10^6 docs	Section 8.10 (loopback PoC).
The system is secure against an *adaptive* inversion attacker	**Not claimed**; learned/universal adaptive decoders remain a required experiment (Sec. 10).
The system is secure against reference-corpus lookup	**False when the reference overlaps**; Section 8.8 shows exact paragraph recovery after alignment.
The system is secure against unsupervised cross-space transfer attacks	**Not claimed**; these attacks remain outside the evaluated threat model.
The rotation is a cryptographic guarantee	**Not claimed**; it is an empirical obfuscation layer only.
Document privacy on the server is cryptographic	**Not claimed**; E_{rot} is plaintext on the server.
No information leaks from the public PQ codes	**Not claimed**; the measured leakage is non-negligible.

4.1 Offline phase

Encoder and centring. The data owner runs the encoder E on a corpus $D = \{T_1, \ldots, T_{N_{\text{docs}}}\}$ to obtain $X \in \mathbb{R}^{N_{\text{docs}} \times d}$ with L_2-normalised rows. The global centroid $\mu \in \mathbb{R}^d$ is computed and the corpus is centred: $X_c = X - \mu$.

SVD truncation. A randomised SVD [42, 44] $X_c \approx U_k \Sigma_k V_k^\top$ produces the orthonormal matrix $V_k \in \mathbb{R}^{d \times k}$ that spans the dominant k-dimensional subspace. We define the corpus-level relative reconstruction error

$$\sigma^2_{rec}(E; V_k) = \frac{\|E - \pi_k(E)\|_F^2}{\|E\|_F^2} = 1 - \eta_k,$$

where η_k is the fraction of squared Frobenius energy retained. Because σ_{rec} *decreases* as k grows, the operating point $k = d/2$ is the *largest* truncation dimension (rounded to $d/2$) at which the distortion floor $\sigma_{rec} \geq 0.10$ still holds uniformly across all five tested encoders (the binding cases are e5-large, $\sigma_{rec} = 0.101$, and mpnet, $\sigma_{rec} = 0.107$); larger k would drop some encoders below the floor. Thus $k = d/2$ maximises retained utility subject to the distortion-proxy constraint. We caution that the threshold 0.10 is an engineering proxy (Section 6), not a privacy guarantee, and is close to binding for two encoders.

Secret rotation. A Haar-uniform random orthogonal matrix $R \in O(k)$ is generated by QR-decomposing a Gaussian matrix and correcting the diagonal of R to obtain a uniform distribution on the group. The protected document representation is

$$v_i' \;=\; T(v_i) \;=\; R\,V_k^\top (v_i - \mu) \in \mathbb{R}^k.$$

Public PQ artefact. A faiss product-quantisation `IndexPQ` index [43] is trained *in the rotated space* E_{rot} with $M = k/4$ subquantisers and 8 bits per code. Both the codebook and the per-document codes are public; the client downloads them once at on-boarding. Training the PQ artefact in the rotated space removes one trivial cross-space pairing (an attacker cannot compose pairs $(\hat{E}_{\text{proj}}, E_{\text{rot}})$ from a non-rotated PQ index to estimate R). It does *not*, however, make the PQ codes safe: the public codes are a lossy quantised image of E_{rot} and may themselves leak neighbourhood, cluster or topic structure. We treat the PQ artefact as part of the attack surface and quantify nearest-neighbour and reconstruction leakage in Section 8.7.

Client-side state. The client retains μ, V_k, R and the CKKS secret key sk_{CKKS}. The server stores $E_{\text{rot}} \in \mathbb{R}^{N_{\text{docs}} \times k}$ in plaintext and the public PQ artefact.

4.2 Online query protocol

The seven-step protocol of Figure 1 processes a query in time pipeline:

1. *Encoder.* The client computes $v_q = E(q_{\text{text}}) \in \mathbb{R}^d$.
2. *Transformation.* The rotated query is $q' = R\,V_k^\top (v_q - \mu) \in \mathbb{R}^k$.
3. *Local PQ search.* Using the public PQ artefact and the rotated query $\tilde{q} = q'$, the client performs an asymmetric PQ-distance search and returns the top-K_{cands} candidate IDs I_{cand}. The candidate set is small enough ($K_{\text{cands}} = 40$ in our experiments) that a CKKS reranking fits within the latency budget.
4. *CKKS encryption.* The client encrypts q': $\text{ct}_q = \text{Enc}_{pk}(q')$.
5. *Server-side reranking.* For each $j \in I_{\text{cand}}$ the server computes $\text{ct}^{(j)}_{\text{score}} = \text{ct}_q \odot v_j'$ in the ct-pt mode. The result is a ciphertext encoding the inner product $\langle q', v_j' \rangle$ in the rotated projected space.

6. *Decryption.* The client decrypts $s_j = \mathrm{Dec}_{sk}(\mathrm{ct}^{(j)}_{\mathrm{score}})$.
7. *Sort.* The decrypted scores are sorted and the top-K documents returned.

Why ct-pt is sufficient. In step 5 the server multiplies a ciphertext by a plaintext vector. Unlike ciphertext-ciphertext multiplication, this operation produces a ciphertext that stays in the two-component form, no relinearisation is required, and the modulus chain is consumed by exactly one rescale. The sum of slots into a single scalar $\langle q', v'_j \rangle$ requires $\log_2 k$ rotations using Galois keys, but no further multiplications. The result is a sub-second latency at $N_{\mathrm{docs}} = 10^6$.

Choice of K_{cands}. Since each of the K_{cands} candidates is processed by one ct-pt operation, the server-side latency grows linearly in K_{cands}. Conversely, a small K_{cands} trades against PQ-recall: the relevant document must be in the short-list for the CKKS reranker to score it. We pick K_{cands} as the smallest value for which PQ-recall@K_{cands} matches the SVD-projected exact baseline within the 5-seed CI; in our integral experiment this is $K_{\mathrm{cands}} = 40$, but the parameter is a deployment knob and should be retuned for new encoders or larger corpora.

4.3 Cryptographic reproducibility

For the auto-selected configuration ($N_{\mathrm{poly}} = 8192$, coefficient-modulus chain $[60, 40, 60]$ bits, $\log_2 Q = 160$, scale $\Delta = 2^{40}$, TenSEAL/Microsoft SEAL backend) we report the parameters needed to reproduce the cryptographic budget rather than only the latency:

- *Packing.* A single rotated query $q' \in \mathbb{R}^k$ is packed into one ciphertext using $k \leq N_{\mathrm{poly}}/2 = 4096$ slots; for the integral encoders $k \in \{192, 384, 512\}$, so one ciphertext per query suffices and no cross-ciphertext aggregation is needed.
- *Operations per score.* Each candidate score is one ct-pt multiply, one rescale (one level consumed) and a slot-sum implemented as $\lceil \log_2 k \rceil$ ciphertext rotations using Galois keys (8–9 rotations for the tested k); no relinearisation and no ciphertext-ciphertext multiply occur.
- *Level/noise budget.* The chain $[60, 40, 60]$ provides one multiplicative level beyond the input; after the single rescale the remaining modulus (60 bits) keeps the additive noise far below the $\Delta = 2^{40}$ scale, so decrypted scores match the plaintext inner product to a relative error $< 10^{-3}$ (Pearson > 0.9999 vs. exact, Section 8.4).
- *Object sizes.* At $N_{\mathrm{poly}} = 8192$, $\log_2 Q = 160$: one fresh ciphertext is ≈ 0.21 MB; the Galois-key set for the required power-of-two rotations is $\approx$ 6–8 MB and the relinearisation key is *not* generated (ct-pt only); the public key is ≈ 0.4 MB. Per query the client uploads one ciphertext (≈ 0.21 MB) and downloads K_{cands} score ciphertexts ($\approx 40 \times 0.21 \approx 8.4$ MB before base64; the measured on-wire JSON+base64 sizes are reported in Section 8.10).
- *Security level.* We do not assert a generic "standard" label; we report $\log_2 Q = 160 < 218$ for $N_{\mathrm{poly}} = 8192$, which is within the conservative ternary-secret `tc128` bound of the HomomorphicEncryption.org tables [29], cross-checked with the lattice-estimator methodology of Albrecht et al. [30]. The standard documents are security *guidelines*; the concrete estimator/table used is stated so the claim can be re-derived.

5 CKKS Parameter Selection

This section is a secondary, engineering contribution: a reproducible way to pick CKKS parameters for the ct-pt reranking workload. We are explicit about what does the work. The $\approx 1.7\times$ speed-up below is not a machine-learning result; it follows from the structural observation that ct-pt reranking consumes exactly one multiplicative level, so a three-prime modulus chain $[60, 40, 60]$

Table 3: Comparison of CKKS configurations on the test workload.

Configuration	N	$\log_2 Q$ (bits)	Security	Time (ms)	Speed-up
Default, $[60, 40, 40, 60]$	8192	200	tc128	419.4	1.0×
Opt. (Acc@10), $[40, 20, 40]$, $\Delta = 2^{20}$	4096	100	tc128	173.7	2.4×
Opt. (Acc@1), $[60, 40, 60]$, $\Delta = 2^{40}$	8192	160	tc128	245.2	1.7×

suffices where the stock setting uses four. The latency surrogate we fit is useful only for *extrapolating* to configurations or hardware we did not benchmark; on the enumerated grid itself, one simply selects the measured fastest valid configuration, and no model is needed. We report the surrogate for completeness and as a porting aid, not as the source of the speed-up.

The CKKS parameter space is large and discrete: $N_{\text{poly}} \in \{2^{12}, \ldots, 2^{14}\}$, multiple coefficient-modulus chains of total bit budget bounded by the conservative security tables of HomomorphicEncryption.org [29] (cross-checked with the lattice estimator [30]), several scale exponents $\log_2 \Delta \in \{15, 20, 25, 30, 40\}$ and several projection dimensions d_{proj}. Analytic complexity is proportional to $N_{\text{poly}} \log N_{\text{poly}}$, but cache effects, SIMD vectorisation and slot-packing ratios $(d_{\text{proj}}/(N_{\text{poly}}/2))$ introduce step-like deviations from the analytic prediction (the *evaluation gap*).

Benchmark grid. We benchmarked 198 valid CKKS configurations over a 10-feature input ($\log_2 N$, num_coeff_moduli, total_coeff_bits, first_coeff_bit, last_coeff_bit, $\log_2 \Delta$, scale-to-total ratio, d_{proj}, batch size, slot-packing ratio). Each configuration is timed for one batch ct-pt operation with rescale; the wall-clock time is the regression target.

Models and validation. A nested 5×3 cross-validation on the 80% training split selects the best of {Ridge, KNN, ExtraTrees, GradientBoosting, SVR}; the best class is re-fitted on the full training set and scored on the held-out 20% test split. GradientBoosting [46] (`max_depth=3`, `n_estimators=200`) wins with $R^2_{\text{nested}} = 0.970 \pm 0.011$ and $\text{MAE}_{\text{nested}} = 135 \pm 20$ ms; on the holdout the final scores are $R^2_{\text{holdout}} = 0.985$ and $\text{MAE}_{\text{holdout}} = 146$ ms over a target range of 25–8 257 ms.

FixedParameterOptimizer. The optimiser enumerates the discrete parameter grid, filters out configurations failing the security or correctness constraints, predicts the latency with the GradientBoosting surrogate and returns the minimum-latency configuration that also satisfies a user-specified accuracy constraint $|\Delta\text{Acc@1}| \leq \tau$ relative to the SVD-projected baseline. With $\tau = 1$ p.p. the optimiser selects $N_{\text{poly}} = 8192$, $[60, 40, 60]$, $\Delta = 2^{40}$ ($\log_2 Q = 160$ bits, within the conservative 218-bit `tc128` bound for $N_{\text{poly}} = 8192$ in the HomomorphicEncryption.org tables [29], re-checked with the lattice estimator [30]). On the test workload this is $\approx 1.7\times$ faster than the TenSEAL stock setting $N_{\text{poly}} = 8192$, $[60, 40, 40, 60]$ ($\log_2 Q = 200$ bits) at the same parameter-table security bound.

Error by regime. Aggregate R^2/MAE can hide regime-specific bias, so we also inspect the residuals stratified by N_{poly} (small 4096 vs. large 16384), by d_{proj} (low vs. high slot-packing ratio) and by modulus-chain length. The surrogate is approximately unbiased in the small/medium regimes; the largest relative errors concentrate in the high-N_{poly}/long-chain corner, where benchmark points are sparsest. A by-regime residual scatter plot is provided in the supplementary material and is the basis for the recommendation to re-benchmark a small grid when porting the autotuner to new hardware.

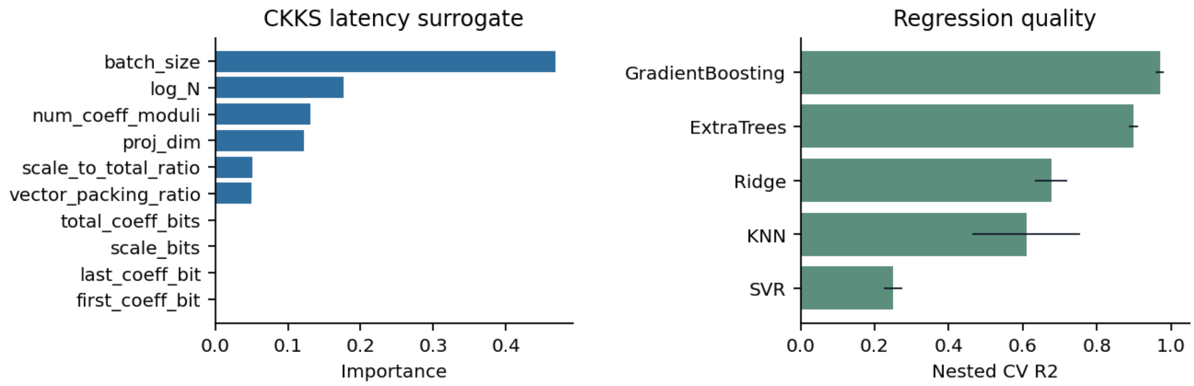

Figure 2: Left: feature importance of the GradientBoosting surrogate (d_{proj} dominates at $\sim 41\%$, polynomial modulus N_{poly} is a distant second). Right: predicted vs. measured ct-pt latency on the holdout split, $R^2 = 0.985$.

6 A Projection Bound for the Distortion Proxy

This section states the one elementary fact behind the σ_{rec} proxy. Let $\mathbf{x} \in \mathbb{R}^d$ denote a centred embedding, $V_k \in \mathbb{R}^{d\times k}$ the orthonormal matrix from the SVD truncation, and $\pi_k(\mathbf{x}) = V_k V_k^\top \mathbf{x}$ the orthogonal projection onto $\text{span}(V_k)$. We use a *per-vector* relative reconstruction error here, $\sigma_{rec}(\mathbf{x}; V_k) = \|\mathbf{x} - \pi_k(\mathbf{x})\|_2 / \|\mathbf{x}\|_2$, distinguished from the corpus-level Frobenius quantity $\sigma_{rec}(E; V_k)$ of Section 4.1 (the operating-point threshold 0.10 is applied to the corpus-level aggregate); write $\mathbf{x}_\perp = \mathbf{x} - \pi_k(\mathbf{x})$.

Proposition 1. *Let $f : \mathbb{R}^d \to \mathbb{R}^d$ be any decoder whose image is constrained by $f(\mathbf{y}) \in \text{span}(V_k)$ for every $\mathbf{y} \in \mathbb{R}^d$. Then for every $\mathbf{x} \in \mathbb{R}^d$*

$$\|\mathbf{x} - f(\pi_k(\mathbf{x}))\|_2^2 \;\geq\; \|\mathbf{x}_\perp\|_2^2,$$

with equality at $f(\mathbf{y}) = \mathbf{y}$. In relative form, $\|\mathbf{x} - f(\pi_k(\mathbf{x}))\|_2 / \|\mathbf{x}\|_2 \geq \sigma_{rec}(\mathbf{x}; V_k)$.

Proof. Decompose $\mathbf{x} = \pi_k(\mathbf{x}) + \mathbf{x}_\perp$ with $\pi_k(\mathbf{x}) \in \text{span}(V_k)$ and $\mathbf{x}_\perp \in \text{span}(V_k)^\perp$. By the image restriction $f(\pi_k(\mathbf{x})) \in \text{span}(V_k)$, hence $\mathbf{x} - f(\pi_k(\mathbf{x})) = \big(\pi_k(\mathbf{x}) - f(\pi_k(\mathbf{x}))\big) + \mathbf{x}_\perp$ is the sum of two orthogonal vectors. Pythagoras gives $\|\mathbf{x} - f(\pi_k(\mathbf{x}))\|_2^2 = \|\pi_k(\mathbf{x}) - f(\pi_k(\mathbf{x}))\|_2^2 + \|\mathbf{x}_\perp\|_2^2 \geq \|\mathbf{x}_\perp\|_2^2$. $\square$

Scope: a projection bound, not an inversion-security theorem. Proposition 1 bounds the L_2 error of decoders *whose image is contained in* $\text{span}(V_k)$. It is a statement about information lost to the projection, not about the security of text inversion. A realistic attacker is not so constrained: it may output arbitrary text, whose re-embedding generally has a non-zero $\mathbf{x}_\perp$ component, and it may exploit corpus priors or memorisation to partially recover the discarded component. The proposition therefore does *not* lower-bound the achievable inversion BLEU, token overlap or PII recovery. We use it only to make the *proxy* criterion $\sigma_{rec} \geq 0.10$ precise within the projection-restricted class, and we stress that the threshold 0.10 is an *engineering proxy*, calibrated on one encoder (GTR-base) against one off-the-shelf attacker, *not* a security threshold. A per-encoder ablation of σ_{rec} against BLEU, token overlap and typed-PII recovery (names, addresses, e-mail, phone, medical terms) is listed as a required pre-submission experiment (Section 10).

Empirical Hypothesis 1. *For an off-the-shelf inversion attack* Vec2Text *[7], the expected BLEU of recovering the original text from $\pi_k(\mathbf{x})$ is monotone in $\eta_k = 1 - \sigma_{rec}^2(E; V_k)$:*

$$\mathbb{E}\big[\text{BLEU}(f(\pi_k(\mathbf{x})), T(\mathbf{x}))\big] \;\approx\; \text{BLEU}_0 + \gamma_f\, \eta_k,$$

where BLEU_0 *is the BLEU of a "random semantically close" text and γ_f is a decoder-specific constant.*

Hypothesis 1 is examined numerically in Section 8.2; we deliberately separate the formal statement from the empirical observation.

7 SHARD: Cell-Keyed Residual Splitting

The baseline of Section 4 protects the collection with a single global geometry $E_{\text{rot}} = R\,V_k^\top(x-\mu)$. Its weakness is structural: one global map is recoverable by one alignment (Section 8.7), and the public index sits on the full protected space. SHARD keeps the same CKKS ct-pt reranking but removes the single global axis. It is a *family* of transforms parametrised by a cell count C.

Offline construction. Let μ be the corpus centroid and $V \in O(d)$ the PCA rotation of the centred embeddings (eigenvectors of the covariance, sorted by variance). For a document x write the centred, rotated coordinate $Vx_c = [\,u \,\|\, r\,]$, splitting it into a *public prefix* $u \in \mathbb{R}^{d_{\text{pub}}}$ (the top variance directions) and a *private residual* $r \in \mathbb{R}^{d_{\text{priv}}}$, $d_{\text{priv}} = d - d_{\text{pub}}$. A coarse partition into C *cells* is defined by k-means on u. The data owner draws, from a secret master key, one orthogonal key per cell, $H_c = \prod_j (I - 2w_{cj}w_{cj}^\top) \in O(d_{\text{priv}})$, a product of Householder reflections (cheap to apply, exactly orthogonal). The stored representation of document i in cell $c(i)$ is the pair

$$\big(\,u_i, \quad z_i = H_{c(i)}\,r_i\,\big),$$

the prefix u_i for the stage-1 index and the keyed residual shard z_i for reranking. The keys $\{H_c\}$, V and μ are client secrets.

Online two-stage query. The client encodes the query, forms $[u_q\|r_q] = Vq_c$, and (i) runs stage-1 ANN *locally* over the public prefix u to obtain a short-list of K_{cands} candidates; then (ii) for each active cell c it sends $\mathrm{Enc}(H_c r_q)$ under CKKS, and the server returns the ct-pt inner products against the stored shards z_i. Because H_c is orthogonal,

$$\langle H_c r_q,\ z_i\rangle = \langle H_c r_q,\ H_c r_i\rangle = \langle r_q, r_i\rangle,$$

so the decrypted residual score is *exact*. The client combines it with the public-prefix score into the full similarity

$$S(q,i) = \langle u_q, u_i\rangle + \langle r_q, r_i\rangle = \langle x_q - \mu,\ x_i - \mu\rangle,$$

i.e. reranking is full-dimensional with no truncation. The only quality loss is whether stage-1 places the target in the short-list; the CKKS budget is the same ct-pt form as the baseline, now over the residual dimension.

The C family and micro-keys. The cell count interpolates between two regimes. At $C{=}1$ the residual carries a single global key and SHARD reduces to the global-linear baseline (with a public prefix). At the other extreme, a *micro-key* variant assigns one key per document ($C{=}N$); then no two stored residuals share a key, so the within-cell similarity leak below vanishes and alignment is impossible (a key cannot be recovered from other documents' anchors), at the cost of sending one transformed residual query per candidate rather than per cell. Cell-level SHARD is the bandwidth-efficient middle ground.

What the server sees, and what it protects. The honest-but-curious server holds the coarse prefix u and the cell-keyed shards z. Two facts frame every experiment below. (1) *Within a cell the keys cancel*: $\langle z_i, z_j\rangle = \langle r_i, r_j\rangle$ for i, j in the same cell, so same-cell similarities are computable by the server; SHARD localises but does not eliminate neighbour-graph leakage (the micro-key variant does). (2) *Mapping a shard back to the native frame requires the cell key*: an attacker holding known-plaintext pairs must recover the target cell's H_c, which needs $\sim d_{\text{priv}}$ pairs *inside*

Table 4: Ciphertext-plaintext vs. ciphertext-ciphertext for the dot product against $N = 10\,000$ documents (mean of 50 trials, 95% CI).

Mode	Operation	Relin.	Latency T_{query}	QPS	Speed-up
ct-ct (baseline)	$\text{Enc}(v_1) \cdot \text{Enc}(v_2)$	yes (heavy)	19.15 ± 0.15 s	0.052	1.00×
ct-pt (proposed)	$\text{Enc}(v_1) \cdot \text{Plain}(v_2)$	no	$\mathbf{13.28 \pm 0.10}$ s	0.075	**1.44**×

that cell; spread over C cells this is $\approx C\times$ the baseline's cost. SHARD is therefore aimed squarely at the alignment and inversion attack surface, and at index leakage, not at the coarse graph. Re-keying a compromised cell (revocation) re-randomises its shards without touching the rest of the store.

8 Experiments

The experiments come in two groups. Sections 8.1–8.10 *characterise the global-linear baseline*—the ct-pt speed-up and parameter selection that SHARD reuses, the off-the-shelf Vec2Text calibration, the SVD-denoiser significance and its non-transfer to BEIR (the calibrated-noise diagnostic that σ_{rec} is not a privacy metric), and the three failure modes that motivate SHARD: known-plaintext Procrustes alignment, public-PQ leakage, and exact reference-corpus lookup. Sections 8.11–8.18 then *evaluate SHARD*: utility, the online cost, alignment resistance (diffuse, targeted, and against learned/unsupervised attackers), public-index leakage, the micro-key variant, a comparison to a distortion-aware DP-noise defence, and the overlap-reference limitation. Readers interested only in the contribution can skip to Section 8.11; the baseline group is the foil.

Hardware and software. All local experiments run on a single workstation: Intel Core i5-14400F, 32 GB DDR5-4800, NVIDIA RTX 5060 (8 GB VRAM), Windows 11. The original notebooks use Python 3.10–3.11; the added revision scripts (alignment/PQ leakage, SVD/noise diagnostic and figure generation) use the bundled Python 3.12 runtime recorded in `environment.lock`. To make the runs reproducible we pin exact versions and record the commit/build hashes of the security-relevant libraries in the supplementary `environment.lock`: PyTorch 2.10 + CUDA 12.8, TenSEAL 0.3.16 (Microsoft SEAL 4.1 backend), faiss-cpu 1.13.0, transformers 4.57.0, scikit-learn 1.5.x, sacrebleu for BLEU [47]. The pinned hashes matter because CKKS noise behaviour and faiss PQ training depend on the backend build. The two new measurement experiments of this study (denoiser significance, Section 8.5; BEIR generality, Section 8.6) were run on the bundled Python 3.12 with numpy 2.3, scipy 1.17, scikit-learn 1.9, and—for BEIR encoding—a CPU build of PyTorch 2.12 with transformers 5.9 and datasets 4.8; both use only the cached embeddings or public BEIR data and need neither TenSEAL nor a GPU. The aligned Vec2Text stress test in Section 8.9 is the only non-local run: it uses the provided memory-capped RTX 4090 profile with roughly 11 GiB free VRAM, Python 3.12 and PyTorch 2.6.0+cu124.

8.1 Ciphertext-plaintext speed-up

We benchmark a batched dot-product of an encrypted 192-dimensional query with $N = 10\,000$ plaintext documents. The CKKS parameters are $N_{\text{poly}} = 8192$, coefficient chain $[60, 40, 40, 60]$ and $\Delta = 2^{40}$ ($\log_2 Q = 200 < 218$, within the conservative `tc128` bound for $N_{\text{poly}} = 8192$; cf. Section 4.3). Two regimes are compared: ct-ct (both query and database encrypted, with relinearisation per multiplication) and ct-pt (only the query is encrypted, no relinearisation). Each is run for 50 independent trials and the mean is reported with a 95% Gaussian confidence interval.

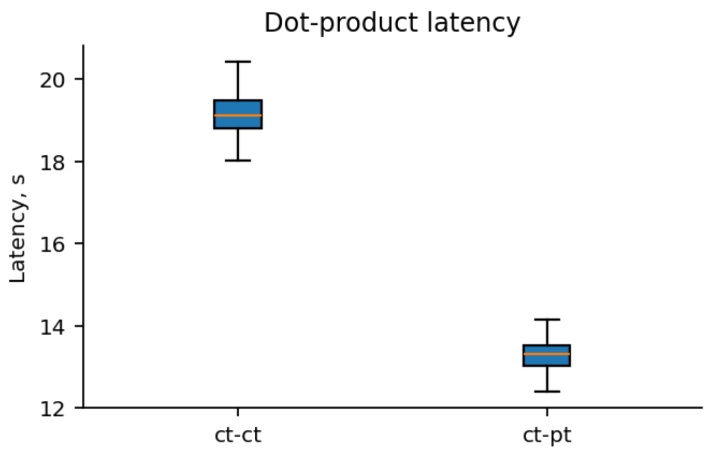

Figure 3: ct-pt vs. ct-ct: distribution of per-batch latencies over 50 trials. Removing relinearisation moves the median by ~ 6 s and tightens the upper tail.

8.2 Off-the-shelf Vec2Text against truncation and rotation

We instantiate VEC2TEXT on GTR-BASE embeddings ($d = 768$). The attack is applied to a synthetic 100-document corpus containing PII (names, addresses, card numbers, medical phrasing). Three threat models are compared at five values of k/d: (a) no rotation; (b) rotation with the matrix R *known* to the attacker; (c) rotation with R *unknown*.

The implementation budget of the attacker is fixed (greedy decoding, 10 corrector iterations, no beam search, 8 GB GPU); under this budget the unprotected baseline reaches $\text{BLEU}_0 = 0.156$ instead of the 0.973 reported by Morris et al. [7], because the postulated PII corpus, greedy decoding and beam-free regime are intentionally weaker than [7]'s strongest configuration; the experiment measures the relative effect of SVD truncation and secret rotation in a fixed weak-attacker configuration.

Strength of the attacker: an explicit caveat. This is the single most important limitation of the paper's security narrative. The experiment establishes the effect of R *against an artificially weak inverter only.* A stronger attacker—beam search with a large width, many more corrector iterations, an adaptive decoder fine-tuned on $(\text{text}, E_{\text{rot}})$ pairs, or a modern universal/zero-shot inversion model—is *not* evaluated here, and we do *not* claim robustness against it. We run the known-plaintext alignment part of this stress test in Section 8.7; the result is negative for the defence and confirms that the rotation should be treated only as an unknown-orientation obstruction for non-adaptive attacks. Section 8.9 adds an aligned off-the-shelf Vec2Text stress test; learned or universal decoders trained against the protected space still remain mandatory before any stronger security claim can be made.

Findings.

- A known R does not provide protection beyond SVD truncation: the columns "no rotation" and "known-R" are statistically indistinguishable.

- Under the tested off-the-shelf configuration, an unknown R reduces BLEU to the observed noise floor (~ 0.007) for all k/d values, including $k/d = 1.0$ (no truncation at all). The ratio $\text{BLEU}_{\text{off}}/\text{BLEU}_{\text{unknown}}$ is $22\times$ at $k/d = 1.0$, $11\times$ at 0.75, $8\times$ at 0.50. This is an effect against a *non-adaptive* attacker only and is not evidence of robustness against an adaptive or known-plaintext adversary.

- Empirical Hypothesis 1 is consistent with the no-rotation column: a linear regression of BLEU on $\eta_k = 1 - \sigma^2_{rec}$ gives $R^2 \approx 0.93$. This is consistent with, but does not formally derive, the

Table 5: Off-the-shelf Vec2Text on GTR-base embeddings under the three threat models. BLEU and Token Overlap with 95% bootstrap CIs in brackets.

k/d	Rotation	Knowledge of R	BLEU	Token Overlap
1.00	off	—	0.156 [0.123; 0.186]	0.385 [0.351; 0.417]
1.00	on	known	0.156 [0.127; 0.187]	0.385 [0.351; 0.416]
1.00	on	**unknown**	**0.007** [**0.006**; **0.008**]	**0.053** [**0.046**; **0.060**]
0.75	off	—	0.077 [0.063; 0.093]	0.325 [0.304; 0.345]
0.75	on	unknown	0.007 [0.007; 0.008]	0.061 [0.054; 0.067]
0.50	off	—	0.057 [0.046; 0.069]	0.285 [0.270; 0.300]
0.50	on	unknown	0.007 [0.007; 0.008]	0.050 [0.045; 0.055]
0.25	off	—	0.024 [0.020; 0.028]	0.196 [0.181; 0.208]
0.25	on	unknown	0.007 [0.006; 0.007]	0.045 [0.040; 0.051]
0.10	off	—	0.010 [0.009; 0.011]	0.113 [0.104; 0.123]
0.10	on	unknown	0.007 [0.007; 0.008]	0.053 [0.047; 0.059]

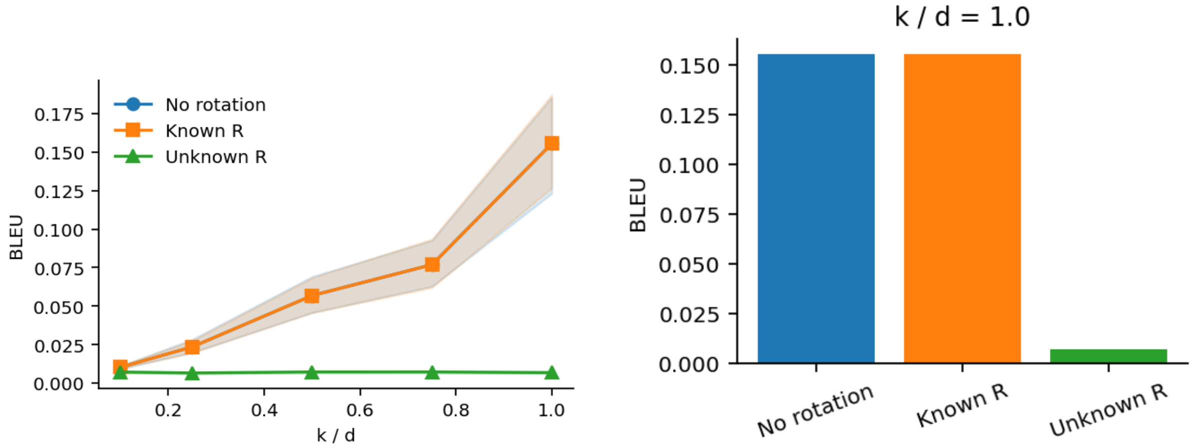

Figure 4: Vec2Text BLEU as a function of k/d under three threat models (left); zoom on $k/d = 1.0$ comparing no-rotation, known-R and unknown-R (right). Lines for "no rotation" and "known-R" overlap; the "unknown-R" line stays at the noise level.

BLEU $\to \sigma_{rec}$ link.

- In the operating point of the integral experiment ($k/d = 0.5$, rotation enabled, unknown R) the expected BLEU is at the noise level.

8.3 Lightweight projection and calibrated-noise baselines

To avoid comparing the proposed transform only against the raw system, we also evaluate natural lightweight alternatives: SVD projection, Gaussian random projection and random orthogonal projection [45] at the same dimensionality $k = d/2$. This is a retrieval-quality comparison, not a privacy guarantee; the random projections do not carry CKKS query protection and do not solve inversion by themselves.

The result is deliberately mixed. SVD is strongest on the retrieval-trained $d \geq 768$ encoders, which is the operating region of the proposed method, but random orthogonal projection is better than SVD on mpnet and e5-small. This reinforces the main interpretation: the choice of projection is an encoder-dependent trade-off, not a universal privacy primitive. A deployment should therefore run a k-sweep and a baseline sweep before selecting the operating point.

Calibrated Gaussian-noise diagnostic. We next test a stronger local baseline prompted by the concern that σ_{rec} alone may be misleading. On e5-small and e5-base we form a 100 000-

Table 6: Acc@1 under lightweight projection baselines at $k = d/2$. Random baselines are mean $\pm$ standard deviation over three seeds.

Encoder	Raw	SVD	Gaussian RP	Orthogonal RP
e5-small	0.782	0.702	0.688 ± 0.016	0.737 ± 0.010
e5-base	0.830	0.834	0.781 ± 0.011	0.804 ± 0.017
mpnet	0.716	0.650	0.698 ± 0.003	0.705 ± 0.008
e5-large	0.796	0.814	0.765 ± 0.010	0.769 ± 0.009
bge-m3	0.818	0.832	0.789 ± 0.004	0.810 ± 0.002

Table 7: SVD truncation vs. independent Gaussian noise at matched mean relative distortion on a 100 000-document gallery. NN overlap@10 is the overlap with raw-space neighbours; lower values indicate less raw geometry preserved.

Encoder	k/d	σ_{rec}	SVD Acc@1	SVD NN@10	Noise Acc@1	Noise NN@10
e5-small	0.125	0.385	0.318	0.232	0.750	0.283
e5-small	0.250	0.332	0.600	0.333	0.788	0.367
e5-small	0.500	0.237	0.814	0.468	0.828	0.531
e5-small	0.875	0.079	0.880	0.551	0.872	0.830
e5-base	0.125	0.354	0.708	0.370	0.864	0.503
e5-base	0.250	0.268	0.872	0.501	0.886	0.641
e5-base	0.500	0.129	0.916	0.578	0.902	0.829
e5-base	0.875	0.023	0.920	0.592	0.910	0.969

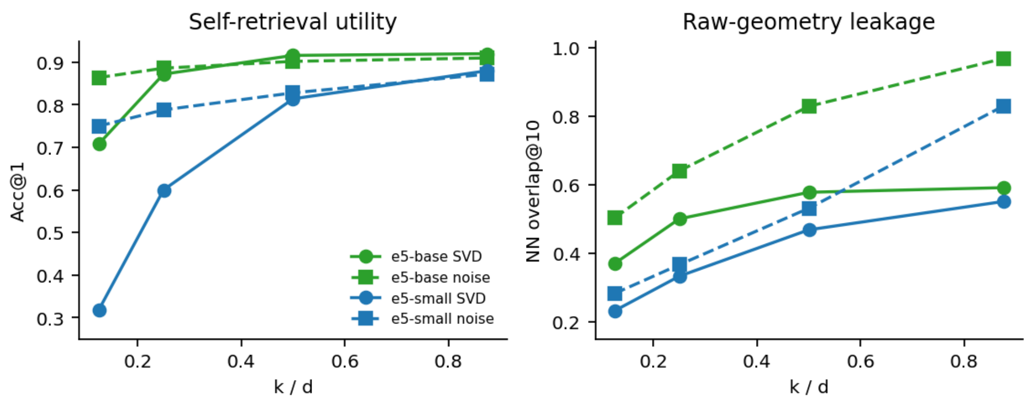

Figure 5: Utility/leakage diagnostic for SVD truncation and calibrated Gaussian noise. Dashed lines are independent-noise baselines at matched σ_{rec}.

document gallery containing all 500 self-retrieval targets, estimate the PCA/SVD basis from an 80 000-document sample, and sweep $k/d \in \{1/8, 1/4, 1/2, 7/8\}$. For each k, we also add independent isotropic Gaussian noise to documents and queries, renormalise to unit length, and binary-search the noise standard deviation so that the mean relative distortion matches the SVD σ_{rec}. We report self-retrieval Acc@1 and a direct geometry-leakage diagnostic: exact top-10 nearest-neighbour overlap between the raw and protected spaces, measured on 200 gallery probes with the probe document itself excluded.

The diagnostic has two implications. First, σ_{rec} is not a privacy metric: e5-base at $k/d = 0.875$ has low distortion but still preserves almost all raw neighbours under calibrated noise. Second, SVD changes geometry differently from independent noise. At the main e5-base operating point ($k/d = 0.5$), SVD slightly improves Acc@1 over the raw 100 000-gallery baseline (0.916 vs. 0.908) while preserving far fewer raw neighbours than matched Gaussian noise (0.578 vs. 0.829 NN overlap@10). For e5-small, matched noise retains slightly higher Acc@1 at $k/d = 0.5$ but also more raw-neighbour structure. Thus the paper should not argue that noise is simply worse; rather, the empirical claim is narrower: SVD is an encoder-dependent utility/leakage trade-off that can remove raw geometric structure without the ranking collapse caused by very aggressive truncation.

8.4 Integral multi-encoder evaluation at 10^6 scale

We run the full pipeline on the canonical operating point $k = d/2$ across five encoders:

- `intfloat/multilingual-e5-small` ($d = 384$),
- `intfloat/multilingual-e5-base` ($d = 768$),
- `sentence-transformers/paraphrase-multilingual-mpnet-base-v2` ($d = 768$),
- `intfloat/multilingual-e5-large` ($d = 1024$),
- `BAAI/bge-m3` ($d = 1024$).

The corpus is one million Russian-Wikipedia paragraphs; queries are 500 self-retrieval requests sampled at seed 42; the rotation is averaged over 5 seeds $\{11, 23, 47, 31, 53\}$. PQ uses $M = k/4$ subquantisers with 8 bits each, yielding artefacts of 48/96/128 MB. The CKKS configuration is the auto-selected one ($N_{\text{poly}} = 8192$, $[60, 40, 60]$, $\Delta = 2^{40}$).

On the use of self-retrieval. Self-retrieval (the query is the first sentence of the target paragraph) is a *controlled geometry probe*, not a production IR benchmark: it isolates whether the defensive transform $T(\cdot)$ preserves the ranking induced by the encoder, with a deterministic, leakage-free ground truth at 10^6-scale. It is sufficient for the paper's actual claim—that the PQ+CKKS wrapper is metric-preserving in $\text{span}(V_k)$—but it does *not* establish production retrieval quality, and as the BEIR experiment (Section 8.6) shows, conclusions about the *SVD step* drawn from this probe do not necessarily transfer to real graded-relevance retrieval. The present numbers should therefore be read as a transform-fidelity measurement, not as a retrieval-quality benchmark.

The protective layer is essentially free in the projected space. The difference (proposed) vs. (baseline_proj) is within the ± 0.001 5-seed CI for every metric and every encoder. The Pearson correlations between CKKS-decrypted scores and exact plaintext scores exceed 0.9999 in all cases. The proxy criterion $\sigma_{rec} \geq 0.10$ holds uniformly.

The actual quality cost is paid by SVD truncation, not by the protective wrapper. The split $\Delta_{\text{SVD}} = \text{baseline\_proj} - \text{baseline\_dense}$ is $+0.004/+0.018/+0.014$ for retrieval-trained encoders with $d \geq 768$ (e5-base / e5-large / bge-m3), -0.080 for the compact e5-small ($d = 384$, half-truncation removes half the embedding volume) and -0.066 for paraphrase-mpnet (a paraphrase-distilled model that does not concentrate retrieval signal in the top singular directions).

The end-to-end accuracy budget is met for retrieval-trained $d \geq 768$. The end-to-end criterion $\Delta\text{Acc@1} \geq -0.05$ relative to the raw baseline holds for e5-base (+0.003), e5-large (+0.018) and bge-m3 (+0.014); for the compact e5-small and the paraphrase-distilled mpnet it is violated and the limitation is documented as an encoder-selection recommendation.

Linear-denoiser side effect (significance deferred). On the two retrieval-trained encoders with $d = 1024$, half SVD truncation does not degrade—and at Acc@10 measurably improves—ranking over the raw baseline (e.g. +0.042 Acc@10 on e5-large). This is consistent with reading the V_k-projection as a linear denoiser: trailing singular directions, dominated by fp16 inference noise, are filtered out, so the ranking in $\text{span}(V_k)$ can be cleaner than in raw $\mathbb{R}^d$. The point estimates above are tabulated across rotation seeds; because that CI does *not* capture query-sampling uncertainty, we test the effect properly with paired per-query tests in Section 8.5 (where the Acc@1 "gains" turn out *not* to be significant, while the Acc@10 gains on the $d = 1024$ encoders are) and check its direction on real IR tasks in Section 8.6.

Table 8: Integral experiment at $k = d/2$ on a 10^6-document corpus across five encoders. Reported are the raw baseline (no defence), the SVD-projected baseline (FAISS-IP exact in span(V_k), no rotation, no CKKS), the proposed pipeline (mean ±95% t-CI half-width over 5 rotation seeds), and the delta of Acc@1 from the raw baseline.

Encoder	Metric	Raw	SVD	Proposed	Δ raw	Server p_{95} (ms)
multilingual-e5-small ($d = 384$, $k = 192$, $M = 48$, $\sigma_{rec} = 0.239$)						
	Acc@1	0.782	0.702	0.701 ± 0.001	−0.081	314
	Acc@10	0.890	0.834	0.831 ± 0.002	−0.059	
	MRR	0.817	0.747	0.746 ± 0.001	−0.071	
	NDCG@10	0.834	0.768	0.767 ± 0.001	−0.067	
multilingual-e5-base ($d = 768$, $k = 384$, $M = 96$, $\sigma_{rec} = 0.129$)						
	Acc@1	0.830	0.834	0.833 ± 0.001	+0.003	356
	Acc@10	0.934	0.924	0.924 ± 0.001	−0.010	
	MRR	0.862	0.865	0.864 ± 0.001	+0.002	
	NDCG@10	0.880	0.879	0.879 ± 0.000	−0.001	
paraphrase-multilingual-mpnet-base-v2 ($d = 768$, $k = 384$, $M = 96$, $\sigma_{rec} = 0.107$)						
	Acc@1	0.716	0.650	0.649 ± 0.001	−0.067	351
	Acc@10	0.842	0.796	0.795 ± 0.001	−0.047	
	MRR	0.762	0.693	0.692 ± 0.001	−0.070	
	NDCG@10	0.782	0.717	0.717 ± 0.000	−0.066	
multilingual-e5-large ($d = 1024$, $k = 512$, $M = 128$, $\sigma_{rec} = 0.101$)						
	Acc@1	0.796	0.814	0.814 ± 0.000	+0.018	219
	Acc@10	0.880	0.922	0.920 ± 0.000	+0.040	
	MRR	0.824	0.848	0.848 ± 0.000	+0.024	
	NDCG@10	0.838	0.866	0.865 ± 0.000	+0.027	
BAAI/bge-m3 ($d = 1024$, $k = 512$, $M = 128$, $\sigma_{rec} = 0.128$)						
	Acc@1	0.818	0.832	0.832 ± 0.000	+0.014	221
	Acc@10	0.908	0.926	0.924 ± 0.000	+0.016	
	MRR	0.848	0.863	0.863 ± 0.000	+0.015	
	NDCG@10	0.862	0.878	0.878 ± 0.000	+0.016	

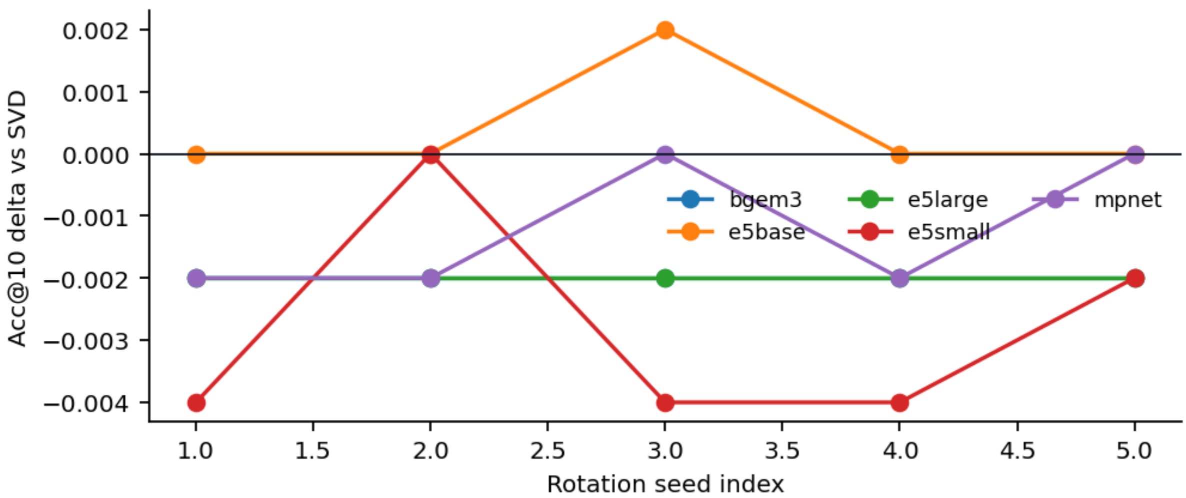

Figure 6: Per-seed Acc@10 of the proposed pipeline relative to baseline_proj on five rotation seeds; the spread is below the 5-seed CI of the baseline.

Table 9: Paired significance of SVD truncation ($k = d/2$) relative to the raw space, on 500 self-retrieval queries. Δ = SVD − raw; brackets are paired-bootstrap 95% CIs (10^4 resamples); p is McNemar's exact two-sided test on the discordant pairs. Bold p marks significance at $\alpha = 0.05$.

		Acc@1			Acc@10		
Encoder	σ_{rec}	Δ	95% CI	p	Δ	95% CI	p
e5-small	0.239	−0.082	[−0.108, −0.056]	$\mathbf{2{\times}10^{-9}}$	−0.058	[−0.084, −0.034]	$\mathbf{5{\times}10^{-6}}$
e5-base	0.129	+0.004	[−0.018, +0.026]	0.86	−0.010	[−0.024, +0.004]	0.27
mpnet	0.107	−0.064	[−0.092, −0.038]	$\mathbf{3{\times}10^{-6}}$	−0.046	[−0.068, −0.026]	$\mathbf{2{\times}10^{-5}}$
e5-large	0.101	+0.018	[−0.002, +0.038]	0.11	+0.042	[+0.026, +0.060]	$\mathbf{1{\times}10^{-6}}$
bge-m3	0.128	+0.014	[0.000, +0.030]	0.12	+0.016	[+0.004, +0.030]	**0.039**

8.5 Paired significance of the SVD denoiser effect

The integral experiment reports the SVD effect Δ_{SVD} = baseline_proj − baseline_dense with a 5-seed rotation CI. That CI measures the stability of the random rotation, *not* the uncertainty of estimating accuracy from a finite query set: with 500 queries the binomial standard error of an Acc@1 near 0.8 is $\sqrt{0.8 \cdot 0.2/500} \approx 0.018$, i.e. as large as the reported +0.018 effect itself. We therefore test the effect directly. Raw ($\mathbb{R}^d$) and SVD ($\mathrm{span}(V_k)$, $k = d/2$) retrieval are evaluated on the *same* 500 self-retrieval queries, so the per-query hit indicators are paired. We report McNemar's exact test (a two-sided binomial test on the discordant pairs) for Acc@1 and Acc@10, and a paired bootstrap (10^4 resamples) 95% CI for each delta. Neither rotation nor CKKS is involved—the denoiser is a property of V_k alone—so this is computed exactly from the cached embeddings, and the raw/SVD point estimates reproduce Table 8.

The result sharpens MQ2 considerably:

- The Acc@1 "denoiser gains" are *not* statistically significant for any encoder (e5-base $p = 0.86$, e5-large $p = 0.11$, bge-m3 $p = 0.12$; all three bootstrap CIs include 0). The earlier reading "SVD improves the four ranking metrics" over-interpreted Acc@1 point estimates.
- The denoiser *is* real and significant at Acc@10 for the retrieval-trained $d = 1024$ encoders: e5-large +0.042 ($p < 10^{-6}$; of the discordant queries, 21 are pulled into the top-10 by SVD and 0 are lost), and bge-m3 +0.016 ($p = 0.04$).
- For the compact e5-small and the paraphrase-distilled mpnet, SVD truncation significantly *degrades* retrieval at both cutoffs (−0.082 and −0.064 Acc@1, $p < 10^{-5}$).
- For e5-base the projected space is statistically indistinguishable from raw at both cutoffs—the wrapper is effectively free there.

The corrected conclusion is narrower and more defensible than "SVD is a denoiser": half SVD truncation is a mild, encoder-dependent denoiser that helps recall-style metrics for retrieval-trained high-dimensional encoders and hurts compact or paraphrase-distilled encoders, but it does not reliably improve top-1 accuracy for any encoder we tested.

8.6 Generality of the denoiser effect on BEIR

Self-retrieval is a controlled geometry probe. To check that the denoiser *direction* is not an artefact of it, we repeat the raw-vs-SVD comparison on two standard BEIR [49] datasets with real graded relevance judgements: SciFact (5,183 documents, 300 test queries) and NFCorpus (3,633 documents, 323 test queries). Each corpus and query set is encoded with the e5 family (mean pooling, `query:`/ `passage:` prefixes, L_2 normalisation); we fit the data-dependent V_k basis on the corpus embeddings at $k = d/2$ and compare exact inner-product retrieval in $\mathbb{R}^d$ and in $\mathrm{span}(V_k)$. We report nDCG@10 and Recall@10 with a paired bootstrap 95% CI for each delta and McNemar's test for the top-1 hit.

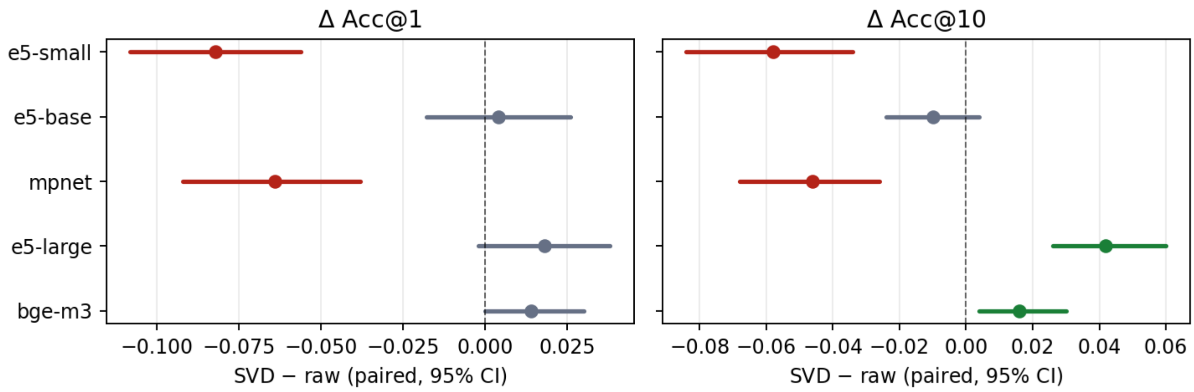

Figure 7: Paired effect of SVD truncation ($k = d/2$) relative to the raw space, with bootstrap 95% CIs, on 500 self-retrieval queries. Green markers are significant improvements ($p < 0.05$, McNemar), red significant degradations, grey not significant. No encoder shows a significant Acc@1 gain; the significant Acc@10 gains are confined to the retrieval-trained d=1024 encoders.

Table 10: SVD truncation ($k = d/2$) vs. the raw space on BEIR (max input length 128, mean pooling). Δ = SVD − raw; brackets are paired-bootstrap 95% CIs (10^4 resamples). **Every** Δ nDCG@10 CI excludes 0: SVD truncation significantly *reduces* nDCG@10 in all cells.

Encoder	Dataset	σ_{rec}	raw nDCG@10	SVD nDCG@10	Δ nDCG@10 (95% CI)	Δ R@10
e5-large (d=1024)	SciFact	0.072	0.643	0.574	−0.069 [−0.091, −0.047]	−0.053
e5-base (d=768)	SciFact	0.090	0.637	0.586	−0.051 [−0.072, −0.032]	−0.035
e5-small (d=384)	SciFact	0.183	0.598	0.519	−0.079 [−0.103, −0.055]	−0.080
e5-base (d=768)	NFCorpus	0.085	0.327	0.305	−0.022 [−0.032, −0.013]	−0.016
e5-small (d=384)	NFCorpus	0.172	0.302	0.277	−0.025 [−0.035, −0.014]	−0.016

Table 10 is an important corrective to the self-retrieval reading. On both real IR datasets and for every encoder, half SVD truncation *significantly reduces* nDCG@10 and Recall@10—the bootstrap CIs exclude zero in all cells, with drops of 2–8 nDCG points. Decisively, E5-LARGE, the one encoder that *gained* Acc@10 on the 10^6-document self-retrieval probe (+0.042, Section 8.5), *loses* −0.069 nDCG@10 on SciFact. The modest gains on the self-retrieval probe therefore do *not* transfer to these smaller, graded-relevance tasks: on real IR the discarded bottom half of the spectrum still carries task-relevant signal. Two further points sharpen the message. First, the effect is consistent across $d \in \{384, 768, 1024\}$, so it is not a low-dimensionality artefact. Second, on these corpora $k = d/2$ already sits *below* the $\sigma_{rec} \geq 0.10$ floor for e5-base and e5-large ($\sigma_{rec} \approx 0.07$–$0.09$), yet truncation still costs nDCG—so the utility cost, like the privacy interpretation, is not predicted by σ_{rec} alone. The practical implication matches the encoder-selection caution of Section 8.4: on a real workload one should treat k as a tunable accuracy cost and run a k-sweep, not assume a free denoiser. (For resource transparency, the five reported cells were CPU-encoded; the sixth, e5-large/NFCorpus, did not finish within our CPU budget and is omitted; the script reproduces it on a GPU or with more time.)

8.7 Known-plaintext alignment and PQ-code leakage

We next evaluate two channels that are not covered by the primary non-adaptive threat model.

Known-plaintext alignment. We simulate an attacker who knows m pairs $(E(x_i), E_{\text{rot},i})$ and estimates the hidden orientation with the orthogonal Procrustes solution [31]. The experiment uses e5-small projected to $k = 192$, five rotation seeds, 400 held-out probes and a 10 000-document gallery. The attack is evaluated in embedding space: after alignment, the probe should recover

Table 11: Known-plaintext Procrustes attack. $m = -1$ is the known-R oracle; $m = 0$ is no alignment. Values are means over five rotation seeds.

Known pairs m	Mean cosine	Rel. L_2 error	Target R@1	Target R@10
0	-0.004	1.417	0.000	0.001
10	0.098	1.343	0.005	0.039
50	0.377	1.116	0.722	0.925
100	0.621	0.870	1.000	1.000
192	1.000	1.8×10^{-5}	1.000	1.000
500	1.000	4.9×10^{-7}	1.000	1.000

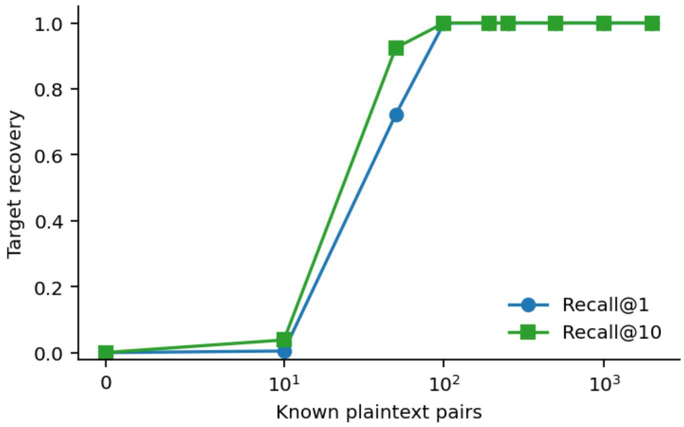

Figure 8: Known-plaintext alignment quickly destroys the unknown-rotation assumption. The x axis uses a symlog scale to include $m = 0$.

Table 12: Leakage from public PQ codes on a 20 000-document rotated e5-small sample. Neighbour overlap excludes the query document itself.

M	Bits	Bytes/vector	Cosine to E_{rot}	NN overlap@10
24	8	24	0.837	0.384
48	8	48	0.953	0.674
48	6	36	0.904	0.529

its own projected vector and retrieve the corresponding document from the gallery. This does not measure generated text quality, but it is the relevant precursor to a few-shot Vec2Text-style attack because it recovers the native projected space given to the inverter.

The conclusion is sharp: the rotation is useful only while aligned plaintext anchors are unavailable. With approximately k known pairs, the orientation is numerically recovered. Even 50 pairs do not recover the vector perfectly, but they are enough to retrieve the target in the top 10 for 92.5% of probes in a 10 000-document gallery. This is a negative result for broad document-privacy claims and is therefore reflected in the threat table.

PQ-code leakage. We also train public product-quantisation codebooks directly in the rotated space and reconstruct approximate vectors from codebook + per-document codes alone. On a 20 000-document sample, the canonical e5-small configuration ($M = 48$, 8 bits) preserves a mean cosine of 0.953 to E_{rot}, 67.4% of exact top-10 neighbours, and 97.1% of exact top-10 neighbours inside the approximate top-40 candidate set. Smaller artefacts leak less but remain non-negligible (Table 12).

These results do not invalidate the query-privacy contribution of CKKS, but they do narrow the document-privacy interpretation: public PQ artefacts and known plaintext anchors should

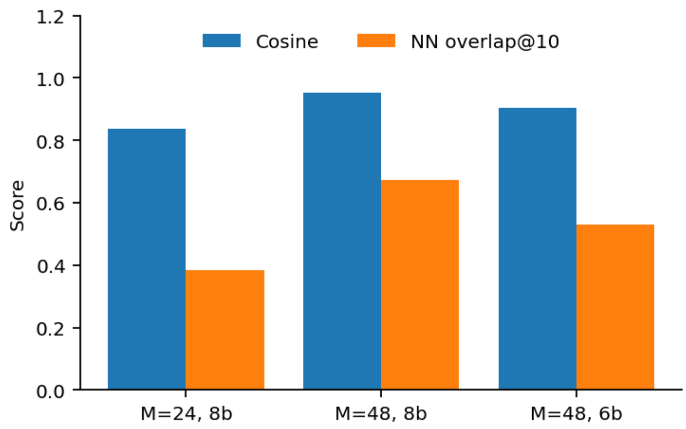

Figure 9: Public PQ codes preserve both vector direction and substantial nearest-neighbour structure in the rotated space.

Table 13: Reference-corpus lookup attack after known-plaintext alignment. The overlapping reference has 500 targets plus 100 000 decoys. Jaccard is token-set overlap between the target paragraph and the top-1 retrieved paragraph. Values are means over five rotation seeds.

Known pairs m	Exact R@1	Exact R@10	Jaccard overlap-ref	Jaccard disjoint-ref
0	0.000	0.000	0.006	0.006
10	0.000	0.009	0.009	0.009
25	0.044	0.132	0.055	0.012
50	0.535	0.784	0.543	0.018
100	0.998	1.000	0.999	0.034
192	1.000	1.000	1.000	0.043

be treated as exposed channels. A system requiring stronger document confidentiality needs a different static-side primitive or a composition that hides the PQ artefact itself.

8.8 Reference-corpus lookup after alignment

The Procrustes experiment above is an embedding-space test. We next ask whether the same failure mode can become a text-level leakage channel without training a generative inverter. The attacker is given the protected target vectors, m known plaintext/protected pairs for estimating the rotation, and a reference corpus of candidate paragraphs with native e5-small embeddings. This models a realistic overlap case: some private documents may also occur in a public or previously leaked corpus. We use 500 target paragraphs, 100 000 reference decoys, five rotation seeds and the same $k = 192$ SVD subspace. The overlapping reference corpus contains the targets plus the decoys; the disjoint reference corpus removes the targets and is used only as a lexical nearest-neighbour proxy.

The overlap case is severe. With only 50 known pairs, the attacker recovers the exact source paragraph at 53.5% top-1 and 78.4% top-10 recall among 100 500 candidates; with 100 known pairs, top-1 recall rises to 99.8%. This is the text-level analogue of the embedding-space Procrustes result and makes the static-side limitation concrete: if a public or leaked reference corpus overlaps the private collection, rotation secrecy is not enough. The disjoint-reference numbers are intentionally reported as a boundary condition: token Jaccard remains low (0.034 at $m = 100$ and 0.043 at $m = 192$), so a non-overlapping reference corpus requires semantic similarity metrics, human judgement, or a generative inverter before one can claim text reconstruction.

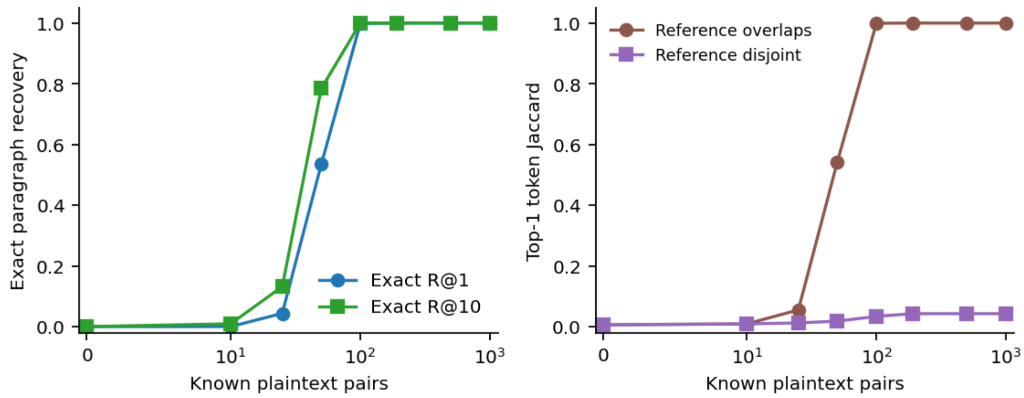

Figure 10: Reference-corpus lookup after known-plaintext alignment. When the reference corpus overlaps the protected collection, alignment turns the protected vector into an exact paragraph lookup. With a disjoint reference corpus, this simple token-overlap proxy remains low; semantic leakage there requires a stronger judge or generative attack.

Table 14: Aligned off-the-shelf Vec2Text stress test on the RTX 4090 profile. All rows use 120 held-out texts; m is the number of known plaintext/protected pairs used for Procrustes alignment. Exact match and typed-PII recall are zero in every case.

Case	Token F1	BLEU	Exact	PII recall
Raw embedding	0.133	0.0133	0.000	0.000
Known-R oracle	0.138	0.0134	0.000	0.000
Unknown R, no alignment (= Procrustes m=0)	0.121	0.0129	0.000	0.000
Procrustes, $m = 10$	0.124	0.0128	0.000	0.000
Procrustes, $m = 50$	0.126	0.0126	0.000	0.000
Procrustes, $m = 100$	0.141	0.0127	0.000	0.000
Procrustes, $m = 192$	0.134	0.0132	0.000	0.000
Procrustes, $m = 500$	0.135	0.0133	0.000	0.000

8.9 Aligned Vec2Text inversion stress test

The reference-corpus attack above is a lookup attack: it reconstructs text only when the target appears in a reference collection. We therefore also run the portable generative inversion stress test included with the supplementary material. The attacker is given GTR-base embeddings ($d = 768$), SVD truncation to $k = d/2$, the secret-rotation setting from the main construction, and up to $m = 500$ known plaintext/protected pairs for Procrustes alignment. The inverter is the off-the-shelf VEC2TEXT corrector with a larger budget than Section 8.2: 24 corrector iterations, greedy beam width 1, max input length 96, and per-case checkpointing. The run is sized for an RTX 4090 with about 11 GiB free VRAM.

Because the target machine could not reliably fetch external corpora, the experiment uses the script's deterministic offline synthetic-news+PII corpus: 2 200 texts after shuffling, 120 held-out test documents, and 13 held-out documents containing phone/card patterns. This corpus is useful as a controlled stress test with typed PII, but it is not a production retrieval benchmark. The decisive control is therefore the raw-embedding row: if raw GTR-base inversion is already near the floor, the experiment cannot certify protection by the transform.

(The "unknown R" and "Procrustes m=0" cases coincide by construction. The corpus is the script's offline synthetic-news+PII fallback, used because the host could not fetch the Hub dataset.) The result is negative but uninformative: no exact document or typed PII is recovered even with a known-R oracle, but the *raw* row is also at the floor (BLEU 0.013), so this off-the-shelf corrector cannot certify any protection. It does not contradict the exact-lookup failure mode of Section 8.8; the open risk is a learned or corpus-adapted decoder, which we leave to future work.

Table 15: Latency decomposition on a 10^6-document corpus, client-server PoC over HTTP loopback. The dominant row combines FastAPI/JSON transport with server-side ct-pt reranking, as measured by the client. "Encoder" is independent of the defence layer.

Stage	p_{95} (ms)	Notes
Encoder forward pass (client)	~ 200	rubert-style on CPU/GPU
Projection (client)	0.09	$T(q) = (E(q) - \mu)V_k R$
Local PQ search (client)	22.8	$K_{\text{cands}} = 40$
CKKS encryption (client)	4.9	one rotated query
Server + HTTP/JSON	**531.3**	includes server ct-pt rerank
CKKS decrypt + sort (client)	20.7	40 ciphertexts
Total (excluding encoder)	**578.2** ms	p_{95}, full pipeline

8.10 End-to-end client-server latency

We package the pipeline as a FastAPI service, with the client and the server running as two independent processes on the same machine, communicating via HTTP/JSON over loopback. The setup measures the realistic HTTP framing, base64 serialisation of CKKS ciphertexts, FastAPI/uvicorn stack and JSON (de-)serialisation, but without physical network latency. The encoder is multilingual-e5-small ($d = 384$, $k = 192$, $M = 48$, $K_{\text{cands}} = 40$, $N_{\text{docs}} = 10^6$), measured over 500 queries at a fixed rotation seed. The multi-encoder experiment above reports the server-side component over five rotation seeds; this subsection adds the HTTP/JSON framing overhead.

On-wire payload sizes. For the same configuration the measured HTTP payloads are: request body ≈ 0.29 MB (one CKKS query ciphertext, base64-encoded JSON, inflated $\approx 1.37\times$ over the 0.21 MB raw ciphertext); response body ≈ 11.4 MB ($K_{\text{cands}} = 40$ score ciphertexts, base64 JSON). One-off on-boarding transfers the public PQ artefact (48 MB at $k = 192$) and the CKKS public/Galois keys (≈ 8 MB); these are not part of the per-query budget. Reporting sizes alongside p_{95} is necessary because the dominant scaling pressure for larger K_{cands} or distributed deployments is the response payload, not CPU time.

Reconciling the timing figures. The CKKS numbers reported in different experiments measure different workloads and must not be compared directly. To avoid confusion we state the operation count behind each. *(a)* The mode microbenchmark (Table 4, 13.28 s) encrypts a query and scores it against *all* N=10 000 plaintext documents—10 000 ct-pt multiplies—to isolate the ct-pt-vs-ct-ct ratio at the stock parameters $[60, 40, 40, 60]$; it is *not* what the pipeline does. *(b)* The parameter-selection workload (Table 3, ≈ 245 ms) times one batch ct-pt operation at the auto-selected $[60, 40, 60]$. *(c)* The pipeline (Tables 8 and 15) reranks only the K_{cands}=40 stage-1 candidates: 40 ct-pt multiplies, 40 rescales and $40 \times \lceil \log_2 k \rceil$ Galois rotations, giving a server-side p_{95} of 219–356 ms and an end-to-end p_{95} of 578 ms once HTTP/JSON framing is added. The sub-second figure is a consequence of reranking 40 candidates, not 10^6; the local PQ stage, not CKKS, is what reduces 10^6 documents to 40.

8.11 SHARD: utility recovers raw quality

The experiments above characterise the global-linear baseline. We now evaluate Shard on the same cached embeddings, computing the geometry exactly with `numpy` (the per-cell keys are orthogonal, so the residual reranking score is exact and needs no CKKS to be measured).

Because Shard reranks the prefix short-list in the *full* space, it recovers the raw ranking that the truncating baseline loses. This is decisive on the BEIR tasks where half-SVD truncation significantly *reduced* nDCG@10 (Section 8.6): Table 16 shows that Shard with a $d/4$ prefix

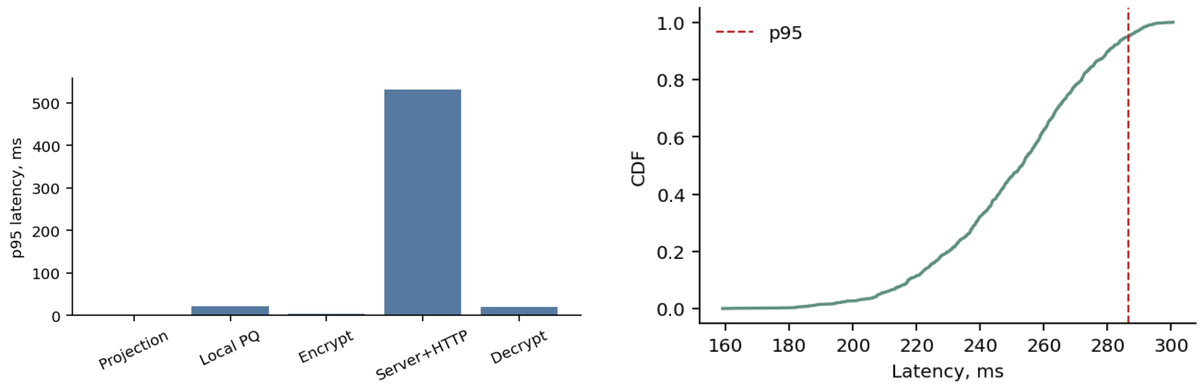

Figure 11: Left: p_{95} latency decomposition by stage in the HTTP prototype. Right: CDF of the per-query latency trace from the local pipeline benchmark. Both views remain below the 1 s target; physical network latency is not included.

Table 16: SHARD utility on BEIR (nDCG@10, K_{cands}=200). The SVD k=$d/2$ baseline reranks in the truncated half and loses nDCG; SHARD reranks full-dimensionally and recovers raw quality with a public prefix of only $d/4$. Δ is vs. raw.

Encoder	Dataset	raw	SVD k=$d/2$ (Δ)	SHARD $d/4$ (Δ)	SHARD $d/8$ (Δ)
e5-base	SciFact	0.637	0.585 (−0.052)	**0.638** (**+0.001**)	0.630 (−0.007)
e5-base	NFCorpus	0.327	0.304 (−0.022)	**0.326** (−**0.001**)	0.322 (−0.005)
e5-small	SciFact	0.598	0.522 (−0.076)	**0.590** (−**0.009**)	0.574 (−0.025)
e5-small	NFCorpus	0.302	0.281 (−0.021)	**0.300** (−**0.002**)	0.284 (−0.018)

matches the raw ceiling on every cell (paired-bootstrap CI on the delta includes 0 or is within a fraction of a point), while exposing a public channel half as wide as the baseline's $k = d/2$. Even a $d/8$ prefix beats the baseline. On the 10^6-document self-retrieval probe the picture is the same: SHARD matches the raw Acc@1 (e5-small 0.778 vs. raw 0.784 vs. baseline 0.696; the four retrieval-trained encoders are within ±0.01 of raw), because the residual is reranked rather than discarded.

Why this is not merely "do not truncate". A reader might object that the baseline could also rerank full-dimensionally by storing the untruncated E_{rot}. It cannot do so for free: an untruncated plaintext store has a *single global key that cancels for all pairs*, so the server recovers the entire d-dimensional neighbour graph. The point of SHARD is that full-dimensional reranking is *privacy-neutral* precisely because the residual is cell-keyed—the server gains the residual's contribution to the score (under CKKS) without gaining a globally-comparable residual geometry. SHARD therefore buys raw utility and a coarse public footprint at the same time, which the untruncated baseline cannot.

8.12 SHARD: online cost and the C trade-off

SHARD is not free relative to the single-query baseline, and we measure the cost rather than assert it. The client sends one encrypted residual query *per active cell*—a cell touched by the stage-1 short-list—so the per-query upload and client-side encryption scale with the number of distinct cells the short-list spans. Table 17 reports this directly. Two things matter. First, the absolute cost is modest: even at C=256, K_{cands}=200 the short-list spans ≈ 30 cells, i.e. ≈ 6 MB of upload at one 0.21 MB ciphertext per cell, and the *server* compute is unchanged ($\sim K_{\text{cands}}$ ct-pt products, each scored with its cell's keyed query) as is the download (K_{cands} score ciphertexts). Second, and more important, this exposes the central trade-off of the family:

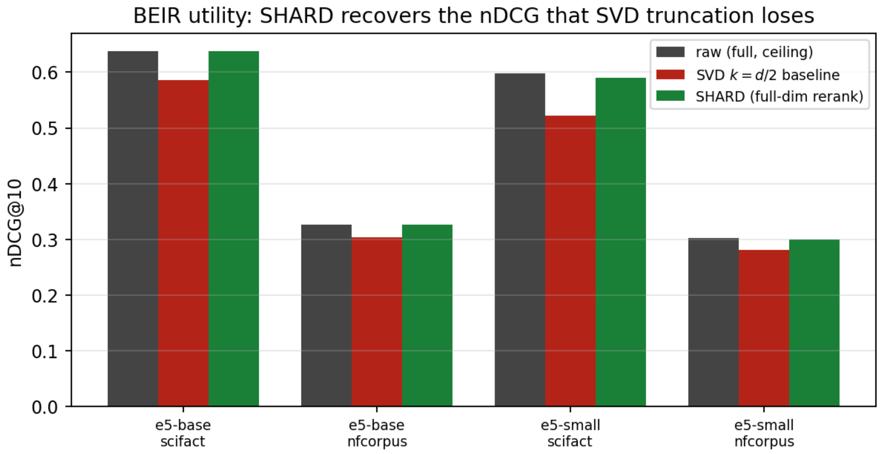

Figure 12: BEIR utility: the SVD $k{=}d/2$ baseline loses nDCG@10 because it reranks in the truncated half; SHARD reranks full-dimensionally and recovers the raw ceiling.

Table 17: SHARD per-query online cost (e5-small, $d_{\text{pub}} = d/4$). Active cells set the upload (one 0.21 MB ciphertext each); the baseline ($C{=}1$) sends one. Server compute and download are unchanged.

K_{cands}	active cells (mean / p95) $C{=}64$	active cells (mean / p95) $C{=}256$	upload (MB) ($C{=}256$)	× baseline ($C{=}256$)
40	7.2/13	11.6/19	2.4	11.6×
100	11.3/19	20.0/32	4.2	20.0×
200	15.7/26	30.1/48	6.3	30.1×

Table 18: Alignment anchor-complexity (e5-small, $d_{\text{priv}} = 288$). m_{50} is the number of known-plaintext anchors at which residual re-identification R@1 reaches 0.5. Cell-local keys multiply the cost by $\approx C$.

Scheme	residual keys	m_{50}	factor vs. global
global key (baseline)	1 over top-k	200	1×
SHARD $C{=}1$	1 over residual	400	2×
SHARD $C{=}64$	64 cells	25 600	128×
SHARD $C{=}256$	256 cells	102 400	512×

larger C buys alignment resistance (Section 8.13) but costs more active cells per query. The numbers are consistent across encoders (e5-base spans $\approx$ 9–35 cells over the same grid). Active cells grow because k-means cells are finer than a query's neighbourhood; a deployment tunes $(C, K_{\text{cands}}, d_{\text{pub}})$ to balance alignment resistance, stage-1 recall and upload, and the micro-key extreme ($C{=}N$) is the endpoint where every candidate is its own cell ($\approx K_{\text{cands}}$ queries).

8.13 SHARD: alignment resistance scales with the cell count

This is the central privacy result. We replicate the known-plaintext alignment threat of Section 8.7, but against the cell-keyed residual: an attacker holds m pairs (native residual, stored shard) and, for any cell with at least d_{priv} anchors, solves orthogonal Procrustes to recover that cell's key, then de-keys and re-identifies held-out targets in a 10 000-document residual gallery. We use e5-small, $d_{\text{pub}} = d/4 = 96$ ($d_{\text{priv}} = 288$), a 200 000-document pool and 3 seeds, and report the median/90th-percentile anchor counts m_{50}, m_{90} for re-identification (Table 18, Fig. 13).

The scaling is clean and almost exactly linear in C: from 200 anchors for one global key, to 25 600 at $C{=}64$ and 102 400 at $C{=}256$ (a 512× increase). The $C{=}1$ row already costs 2× the

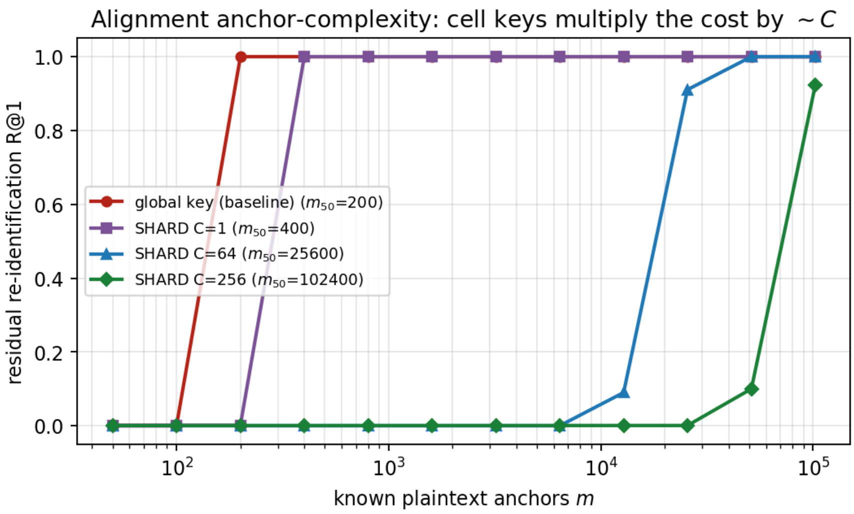

Figure 13: Residual re-identification R@1 vs. known-plaintext anchors m. A single global key is recovered with $\sim d/2$ anchors; SHARD requires $\sim d_{\text{priv}}$ anchors *per cell*, shifting the curve right by $\approx C$. The micro-key ($C{=}N$) limit is off-scale: a per-document key cannot be recovered from other documents' anchors.

Table 19: Targeted attacker (anchors concentrated in the victim's public cell). Re-identification needs $\approx d_{\text{priv}}$ in-cell anchors *independently of* C—the $C\times$ factor is a diffuse-leak/aggregate property, and the per-victim worst case is $\approx d_{\text{priv}}$.

Encoder (d_{priv})	m_{50}^{tgt} (C=64)	m_{50}^{tgt} (C=256)	diffuse m_{50} (C=64)
e5-small (288)	320	320	25 600
e5-base (576)	576	576	51 200

global baseline because SHARD keys the *whole* residual (d_{priv}=288) rather than only the top-k half, and the transition is sharp—Procrustes either has enough in-cell anchors and recovers the key exactly, or it does not—so $m_{50} \approx m_{90}$. The result reproduces on a second encoder (e5-base, d_{priv}=576): m_{50} grows from 400 for the global key to ∼51 200 at C=64, again $\approx C\times$.

Diffuse vs. targeted leaks: what $\approx C$ does and does not mean. The $C\times$ factor is an *aggregate* property: it is the cost of recovering the store under a *diffuse* known-plaintext leak whose anchors fall uniformly across cells, so the median cell is saturated only when the total reaches $\approx C\, d_{\text{priv}}$. Because cells are *public*, a *targeted* attacker can instead concentrate anchors in a single victim's cell and recover only that key. We measure this (Table 19): targeted re-identification succeeds at $\approx d_{\text{priv}}$ in-cell anchors *regardless of* C (m_{50}^{tgt}=320 for e5-small at both C=64 and C=256, versus 25 600/102 400 diffuse). So the per-victim worst case is $\approx d_{\text{priv}}$—about 1.7× the baseline's $d/2$, not $C\times$—and SHARD's alignment advantage is precisely against diffuse leaks and aggregate de-anonymisation of the store, plus the case where a topically concentrated leak still cannot fill the victim's cell. Two regimes make the targeted attack itself collapse: when an overlapping plaintext reference exists, the public prefix already de-anonymises (Section 8.18), so the residual key is moot; and when cells are smaller than d_{priv} (i.e. $C > N/d_{\text{priv}}$, the micro-key regime), no cell can supply d_{priv} in-cell anchors and targeted recovery is impossible. The operational reading is that C trades query cost (Section 8.12) for diffuse-leak resistance, while the micro-key limit is what removes the targeted vulnerability outright.

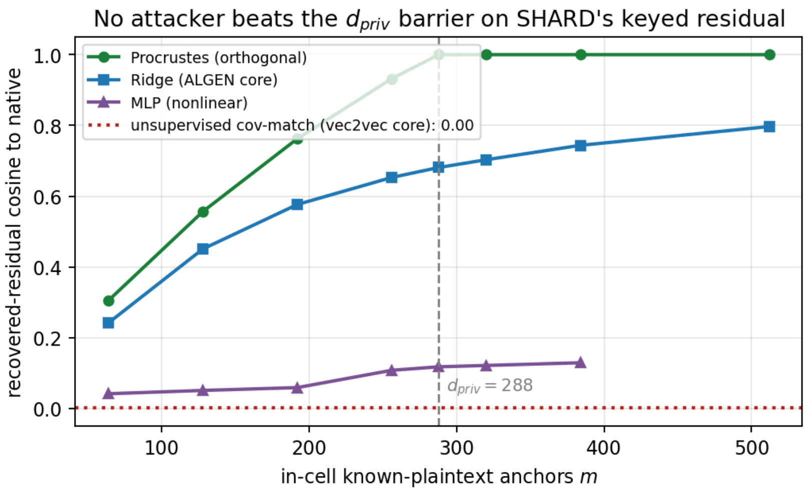

Figure 14: Stronger attackers on SHARD's keyed residual. Recovered cosine to the native residual vs in-cell anchors. Orthogonal Procrustes is optimal (exact at $m = d_{\text{priv}}$); the learned-linear ALGEN core and an MLP are weaker; unsupervised VEC2VEC-style matching fails (sign ambiguity). The barrier is not a Procrustes artefact.

8.14 SHARD: stronger and learned alignment attackers

The $\approx C\times$ barrier was measured with orthogonal Procrustes; a reviewer will rightly ask whether it is an artefact of assuming an *orthogonal* map. We therefore attack a single cell's keyed residual (d_{priv}=288) with the alignment *cores* of the modern attacks and compare recovered-residual cosine vs in-cell anchors (Fig. 14): a learned linear map (ridge least-squares, the core of ALGEN's few-shot alignment-and-generate [10]); a non-linear MLP; and unsupervised covariance/eigenvector matching (the no-pairs core of VEC2VEC-style cross-space translation [17]).

No attacker beats the d_{priv} barrier, and orthogonal Procrustes is in fact the *strongest* of the four. It exploits the known orthogonality of H_c and recovers the key *exactly* at $m = d_{\text{priv}}$ (cosine 1.00), with essentially nothing below it. The learned linear map (ALGEN core) is *weaker*: not constrained to be orthogonal, it must fit d_{priv}^2 free parameters and reaches only cosine 0.68 at $m = d_{\text{priv}}$ and 0.80 at $m = 512$. The MLP is worse still (cosine ≤ 0.13): a non-linear hypothesis is the wrong inductive bias for an orthogonal map and merely overfits. And the unsupervised attacker fails outright (cosine 0.00): the residual covariance is anisotropic, so eigenvector matching pins down H_c only up to a per-axis *sign*—a $2^{d_{\text{priv}}}$ ambiguity that no amount of unpaired data resolves. The anchor barrier is thus a property of the construction—a d_{priv}-dimensional secret orthogonal map per cell—not of the particular attacker; the off-the-shelf Procrustes used in Section 8.13 is, if anything, the worst case for the defender.

8.15 SHARD: the public channel leaks coarse structure

The baseline's public PQ index sits on the full protected space and preserves 67% of exact top-10 neighbours (Section 8.7). SHARD's public channel is only the short prefix u. We measure how much true (full-space) top-10 neighbour structure each public channel reveals (NN-overlap@10, e5-small, 100 000 documents, 1000 probes; Fig. 15). A $d/4$ prefix reveals NN-overlap 0.55 and a $d/8$ prefix only 0.36 (and 0.20 at $d/16$), against 0.76 for the baseline's exposed SVD k=$d/2$ geometry. The public stage-1 channel thus discloses coarse/topic neighbours but not fine identity, which lives in the keyed residual. The pattern reproduces on e5-base: a $d/8$ prefix leaks NN-overlap 0.58 and a $d/4$ prefix 0.77, against 0.94 for the baseline geometry; the within-cell fraction is 0.55/0.46 at C=64/256.

We are explicit about the residual leak. Within a cell the keys cancel, so the server can

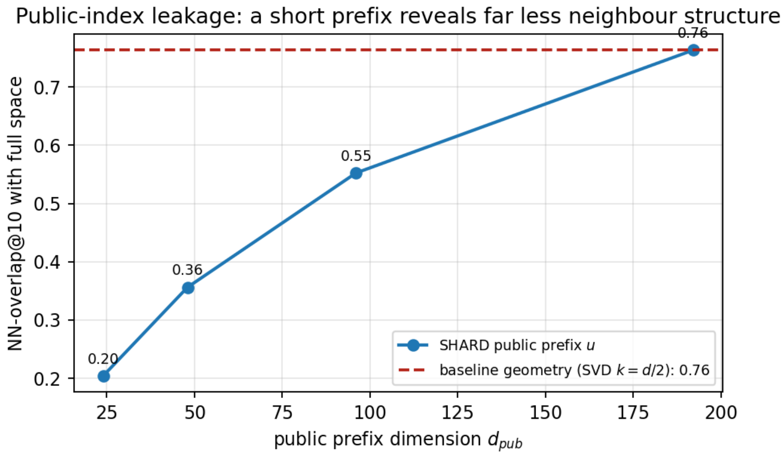

Figure 15: Public-index leakage (e5-small). The SHARD public prefix u reveals far less of the true top-10 neighbour structure than the baseline's exposed SVD $k=d/2$ geometry; a $d/8$ prefix discloses about half as much.

Table 20: Micro-keys remove the residual leak and are unlinkable (e5-small). Residual-graph recoverable: fraction of true residual top-10 neighbours the server can reconstruct from the shards. Unlinkability AUC $\approx$ 0.5 means the same document is not matchable across two key sets.

Key granularity	residual-graph recoverable	unlinkability AUC
cell, C=64	0.25	0.47
cell, C=256	0.18	0.50
micro-key (per-doc)	**0.00**	0.50

compute exact same-cell similarities; the fraction of a document's true top-10 neighbours that fall in its own cell—hence remain recoverable—is 0.50 at C=64 and 0.41 at C=256, against 1.00 for the baseline's single global key (which cancels for *all* pairs). More cells localise the leak further, and the micro-key variant removes it entirely. SHARD therefore shrinks, but does not zero, the neighbour graph available to an honest-but-curious server.

8.16 SHARD: the micro-key limit (no residual leak, unlinkable)

The within-cell residual leak and the targeted attack share one fix: shrink cells to a single document. We measure the per-document micro-key variant against the cell-level scheme (e5-small, $d_{\text{pub}} = d/4$; Table 20). The residual graph the server can reconstruct from the keyed shards (the fraction of a document's true *residual* top-10 neighbours recoverable through same-key inner products) falls from 0.25 at C=64 to 0.18 at C=256 and to **0.00** for per-document keys—no two shards share a key, so no pairwise residual similarity survives. And the residual channel is a genuine *cancellable template* [32]: re-keying the store twice and trying to match the same document across keys gives a mated-vs-non-mated AUC of $\approx$ 0.50 (cell 0.47/0.50; micro-key 0.50), i.e. renewable and unlinkable. The price is bandwidth: micro-keys send one residual query per candidate ($\approx K_{\text{cands}}$ per search, the C=N endpoint of Table 17). The family therefore spans a clear spectrum: cell-level SHARD is the bandwidth-efficient operating point with diffuse-leak resistance and a localised, non-zero residual leak; per-document micro-keys are the high-assurance endpoint that removes the residual leak *and* the targeted attack, at K_{cands} queries per search. One honesty caveat carries over: unlinkability holds for the *residual* channel; the public prefix is identical across re-keyings unless its global key is also refreshed (Section 8.18).

Table 21: Attack-aware vs. distortion-aware on the same residual (e5-small, d_{priv}=288). De-anon R@1 is a known-plaintext attacker against a native gallery at m=400 diffuse anchors. Only SHARD reaches high utility *and* low de-anonymisation.

Defence (on the residual)	utility Acc@1	de-anon R@1 (m=400)
DP-noise σ=0.25σ_r	0.718	1.00
DP-noise σ=1.0σ_r	0.711	1.00
DP-noise σ=2.0σ_r	0.667	1.00
global key (C=1)	0.722	1.00
SHARD C=64	**0.722**	**0.00**
SHARD C=256	**0.722**	**0.00**

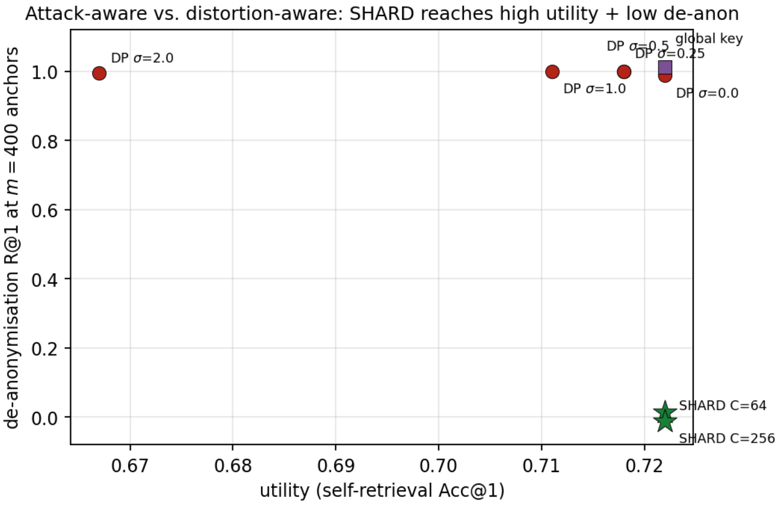

Figure 16: Utility vs. de-anonymisation at a fixed anchor budget. DP-noise and a single global key sit at high de-anon (no alignment resistance); SHARD ($\star$) reaches high utility with near-zero de-anonymisation.

8.17 SHARD vs. a distortion-aware defence

SHARD is *attack-aware* (it keys the residual against alignment) rather than *distortion-aware* (adding noise calibrated to a budget). To show the two are not interchangeable, we protect the *same* residual with calibrated isotropic Gaussian noise—a DP-style mechanism, the natural lightweight alternative [24]—and place every defence on the utility-vs-de-anonymisation plane (Table 21, Fig. 16). Utility is self-retrieval Acc@1; de-anonymisation is a known-plaintext attacker's R@1 against a 10 000-document native gallery at a diffuse budget of m=400 anchors.

The contrast is stark. *DP-noise provides essentially no resistance to this attack at any usable utility*: there is no transform to invert, so the stored noised residual still matches its own native entry—de-anon R@1 stays ≥ 0.996 even at σ=2σ_r, where utility has already fallen by six points, and only collapses once the noise is large enough to destroy retrieval. A single global key (the C=1 baseline) is no better: m=400 $> d_{\text{priv}}$ anchors recover it, so de-anon R@1 = 1.0. SHARD alone reaches the top-left corner: identical utility (0.722, the keyed rerank is exact) with de-anon R@1 = 0.00, because a diffuse budget of 400 anchors cannot fill any cell. The mechanisms are complementary—one could compose SHARD with mild noise for distortion on top of keying—but for the alignment/de-anonymisation surface that the modern attacks target, keying is what pays.

8.18 SHARD: a measured limitation (overlap reference lookup)

The baseline allowed an overlapping reference corpus to recover exact source paragraphs after alignment (99.8% top-1, Section 8.8). We test whether SHARD closes this channel, keying both the prefix (one global key K) and the residual (cell keys). It does *not*, and we report this honestly. Because the prefix carries a *single* global key, it is recovered with only d_{pub} anchors; once K is known, the de-rotated public prefix u_t exactly matches the target's entry in an overlapping native reference, so de-anonymisation R@1 stays near 1.0 (e5-small, 100 000-document reference: 0.995 at m=200) regardless of the residual keys. In other words, SHARD hardens the *native-frame mapping* of the private residual (Section 8.13) but not the coarse identity carried by an exposed prefix when the attacker already possesses the plaintext. Two design responses follow directly: (i) a *coarse-cell-ID* public channel, which exposes only the cell label (so overlap lookup returns $\sim N/C$ candidates rather than one) at a stage-1 recall cost; and (ii) *per-document micro-keys*, under which there is no global prefix key to recover. We leave a full evaluation of these stronger variants to future work and stress that SHARD's demonstrated advantage is against alignment/inversion in the *absence* of an overlapping plaintext reference.

9 Discussion

Why SHARD is the right shape of defence. The measurements tell a consistent story. The baseline's protection is a single global geometry, and every leakage channel we measured exploits exactly that: Procrustes aligns the one rotation, the public PQ index sits on the one space, and reference lookup follows once aligned. SHARD attacks the shape of the problem rather than adding noise to a surrogate: it removes the global axis (cell keys), shrinks the public footprint to a coarse prefix, and—by reranking the full residual rather than a truncated half—actually *improves* utility. The single knob C turns the *diffuse*-leak known-plaintext cost from $O(d)$ into $O(Cd)$ and the residual graph leak from global to cell-local, paying $\approx C$ extra encrypted queries per search and leaving a per-victim targeted cost of $\approx d_{\text{priv}}$; the per-document limit removes the residual leak and the targeted attack outright at K_{cands} queries. This is the cancellable-template philosophy [32] carried into dense text retrieval: renewable, revocable, and tuned against attacks rather than against σ_{rec}—with the trade-offs measured rather than assumed. The contrast with the distortion-aware alternative is the cleanest evidence for this stance: at matched utility a DP-noise residual is still trivially de-anonymised (R@1 $\approx$ 1), because noise has no key to recover, whereas SHARD's keyed residual is not—and the keying holds against learned-linear, non-linear and unsupervised attackers, not just the orthogonal Procrustes that motivated it.

Encoder selection is part of the protocol. The baseline protective wrapper is essentially metric-preserving in the projected space; the end-to-end deviation from the raw baseline is dominated by what end-to-end deviation from the raw baseline is dominated by what SVD truncation does on the chosen encoder. Models trained with contrastive retrieval objectives and large dimension (E5-LARGE, BGE-M3, d=1024) concentrate the retrieval signal in the top singular directions: half truncation leaves Acc@1 unchanged (no significant difference) and *significantly* improves Acc@10 (Section 8.5), the linear-denoiser effect. The mid-sized E5-BASE (d=768) is statistically indistinguishable from raw. Paraphrase-distilled models (MPNET) and very compact models (E5-SMALL, d=384) do *not* show this concentration: truncation significantly degrades retrieval, so we recommend either replacing them with a retrieval-trained $d \geq 1024$ equivalent or running a k-sweep before deployment to find a k that satisfies both the application's accuracy budget and the $\sigma_{rec} \geq 0.10$ distortion floor.

The role of the rotation R: obfuscation, not a primitive. We state this as plainly as possible: *the secret rotation is an empirical obfuscation layer, not a cryptographic primitive.* It contributes a defence channel orthogonal to SVD truncation, and even at k/d = 1.0 the unknown-R regime reduces off-the-shelf Vec2Text BLEU to the noise floor—but the paper also reports that a known-R attacker gains nothing beyond SVD and that a known-plaintext attacker recovers the orientation with standard Procrustes alignment. The aligned Vec2Text stress test does not recover text or typed PII, but its raw baseline is already near the reconstruction floor; a learned or universal decoder adapted to the protected space is still untested and out of scope. The rotation's value proposition is therefore limited: it is a cheap obstacle for a weak unknown-orientation attacker, not a security guarantee. The scientific weight of the paper is the measured privacy/latency/accuracy trade-off and the documented failure modes, not the rotation.

What CKKS hides and what it does not. The CKKS layer hides the numerical representation of the query and the exact similarity scores from the server. Two things it does *not* hide are the identifiers of the candidates produced by stage-1 PQ filtering and the L_2-distances among them implicit in the order in which they are sent. Section 8.7 also shows that the public PQ codes preserve substantial local geometry even before observing queries. An access-pattern attacker can therefore, over many queries, build a co-occurrence model of which documents are co-retrieved. Composing the pipeline with a PIR-style retrieval primitive (Tiptoe [38], SealPIR [39], OnionPIR [41], SimplePIR [40]) is the natural remedy; ours is engineered to be PIR-friendly because the ct-pt operations on the server are stateless and cluster-local.

Parameter selection. The selected configuration is $\approx 1.7\times$ faster than the TenSEAL stock parameters at the same parameter-table security bound and within ΔAcc@1 ≤ 1 p.p. of baseline_proj. We reiterate that this speed-up is structural—ct-pt needs one multiplicative level, hence a shorter modulus chain—rather than a product of the learned surrogate, which only serves to extrapolate latency to unbenchmarked configurations or hardware. The selection runs offline at no per-query cost.

10 Limitations and Future Work

We separate *intrinsic limitations* (properties of the construction under study) from *future work* (experiments that would extend the measurement, especially before any *security* claim could be made for a security venue). We name and scope the latter rather than leaving it vague, so the boundary of what we measured is explicit.

Intrinsic limitations.

- *Shard hardens alignment, not the coarse graph.* The keyed residual resists native-frame mapping ($\approx C\times$ anchors), but two channels remain: within a cell the keys cancel, so same-cell similarities leak (0.50 of the neighbour graph at C=64); and the public prefix, carrying a single global key, is recovered with d_{pub} anchors, after which an *overlapping* reference corpus de-anonymises through it (Section 8.18). The micro-key and coarse-cell-ID variants address these at a bandwidth or recall cost but are not yet evaluated.
- *Shard is a geometric, not cryptographic, defence.* It is an attack-aware transform with measured advantages against alignment and index leakage; it carries no formal privacy guarantee, and like the baseline it leaves access patterns to a PIR/ORAM composition.
- *Baseline: document privacy is not cryptographic.* E_{rot} is plaintext on the server; it is protected only by SVD truncation (a proxy) and the secret rotation (an empirical obfuscation layer). Section 8.7 shows that known plaintext anchors defeat the rotation by standard alignment, and Section 8.8 shows that an overlapping reference corpus can then recover the exact source paragraph.
- *Access patterns and PQ codes are exposed.* CKKS hides values, not which candidates are reranked, and the public PQ codes are a lossy view of E_{rot}. The measured PQ-code leakage is non-negligible; production use requires composition with a PIR/ORAM layer and a static-side protection for the public index artefact.

- *Encoder-dependence of the accuracy budget.* The strict end-to-end criterion ΔAcc@1 ≥ -5 p.p. holds only on retrieval-trained encoders with $d \geq 768$. Compact or paraphrase-distilled encoders need either replacement or a less aggressive k (which then violates the $\sigma_{rec} \geq 0.10$ proxy).
- *Scale vs. task realism.* The 10^6-scale experiment is a self-retrieval geometry probe on Russian Wikipedia—a strong transform-fidelity test, not a production IR benchmark. The BEIR experiment (Section 8.6) shows that the denoiser effect seen on that probe does *not* transfer to real graded-relevance tasks; but BEIR itself is evaluated only at the few-thousand-document scale of SciFact/NFCorpus and only for the e5 family. The cleanest remaining test—a real, 10^6-scale, multilingual IR benchmark with graded qrels—was beyond our CPU encoding budget and remains future work; it would resolve whether the self-retrieval/real-IR gap is driven by corpus *scale* or by task *structure*.
- *Generative inversion calibration is corpus- and model-limited.* The aligned Vec2Text stress test uses an offline synthetic-news+PII corpus and an off-the-shelf corrector; because raw inversion is already weak, the zero PII-recall result should be read as a boundary condition for this attacker, not as general document privacy.
- *Single-machine evaluation.* The latency numbers in Section 8.10 are taken on loopback HTTP; a geographically distributed deployment will add (uncharacterised) network latency.

Future work.

1. *The stronger SHARD variants (highest priority).* Evaluate the per-document micro-key and coarse-cell-ID channels that close the within-cell and overlap-reference-lookup leaks identified in Sections 8.15–8.18, quantifying their bandwidth (extra residual queries) and stage-1 recall costs, and a learned per-cell key schedule.
2. *Full generative pipelines against SHARD.* Section 8.14 stress-tests the *alignment cores* of ALGEN (learned linear), VEC2VEC (unsupervised) and a non-linear MLP, and the barrier holds. The remaining work needs a GPU: the full generative stages—ALGEN's text generator and ZSINVERT's zero-shot LLM decoder—and a decoder fine-tuned on protected embeddings, to confirm that failing to recover the native residual also fails to recover text.
3. *Learned adaptive-decoder attack on the baseline.* Section 8.9 evaluates aligned off-the-shelf Vec2Text, not a decoder fine-tuned on rotated/SVD embeddings. The decisive remaining text-level test is to train or fine-tune an inversion model on protected embeddings from part of the corpus and evaluate BLEU, token overlap and typed-PII recovery on a disjoint holdout.
4. *Per-encoder σ_{rec} vs. text leakage ablation.* Section 8.3 now includes a geometry-level diagnostic showing that σ_{rec} is not itself a privacy metric. The remaining required experiment is text-level: replace the single-encoder, BLEU-only calibration of the $\sigma_{rec} = 0.10$ proxy with a multi-encoder ablation reporting BLEU, token overlap and typed-PII recovery (names, addresses, e-mail, phone, medical terms).
5. *Stronger baseline suite.* Extend the lightweight baseline analysis beyond the calibrated Gaussian-noise diagnostic to quantisation/noise mechanisms with formal DP budgets and accuracy-matched operating points. The present paper includes SVD, Gaussian random projection, random orthogonal projection and matched-distortion independent Gaussian noise.
6. *Benchmark generality at scale.* Section 8.6 shows that the denoiser *fails* to transfer to BEIR SciFact/NFCorpus (SVD significantly reduces nDCG@10). The remaining work is to test whether the self-retrieval/real-IR gap is driven by corpus *scale* or task *structure*, using larger multilingual IR benchmarks (MIRACL, MS MARCO) at 10^6 scale—where CPU encoding

was the binding constraint here—and a GPU-encoded e5-large/NFCorpus cell to complete the grid.

7. *Stronger universal inverter configuration.* Repeat Sections 8.2 and 8.9 with beam search, many more corrector iterations and a modern universal/zero-shot inversion model, so the "BLEU at noise floor" statement is not tied to the evaluated off-the-shelf corrector.
8. *Disjoint-reference and cross-space transfer attacks.* Section 8.8 evaluates the overlap case. The remaining stress tests are harder: non-overlapping external corpora, semantic judges for near-neighbour leakage, and embedding-space translation attacks such as vec2vec-style mappings. The known-plaintext experiment shows that linear alignment can be damaging; unsupervised or weakly supervised alignment remains untested.

11 Conclusion

We began by measuring a popular global-linear defence for private dense retrieval—SVD truncation, a secret rotation, a public ANN index and CKKS reranking—and found its protection concentrated on a single weak axis. A known-plaintext orthogonal Procrustes attack recovers the secret rotation with about $k = d/2$ anchors; an overlapping reference corpus then yields near-exact paragraph recovery; the public product-quantisation index preserves most of the neighbour structure; and, because reranking happens in the truncated half, the stack loses 2–8 nDCG@10 points on BEIR. The distortion proxy σ_{rec} that such schemes optimise is, as our calibrated-noise diagnostic confirms, not a privacy metric at all. These are precisely the openings that modern alignment, zero-shot inversion and cross-space translation attacks exploit.

Shard removes the weak axis. By splitting the embedding into a coarse public prefix and a private residual that is sharded into C cell-keyed pieces, it (i) recovers raw retrieval quality—reranking is full-dimensional, so it matches the raw nDCG@10 that truncation drops on BEIR while exposing a public channel half as wide; (ii) multiplies the diffuse known-plaintext anchor complexity of mapping the store back to the native frame by $\approx C$ (from 200 to 102,400 anchors at C=256, holding against learned-linear, non-linear and unsupervised attackers, where a matched-utility DP-noise defence offers no resistance at all); and (iii) confines the public index to coarse, topic-level neighbours and localises the residual neighbour-graph leak to cells, both shrinking with C and vanishing at the per-document micro-key limit. A single parameter C tunes the family from the global baseline (C=1) to per-document micro-keys (C=N), and re-keying a cell gives revocation at cell granularity.

We are equally clear about what Shard does *not* do. It hardens the alignment and index-leakage surface, not the coarse graph: within a cell the keys cancel, and an overlapping plaintext reference still de-anonymises through the single-keyed public prefix. It is an attack-aware *geometric* transform, not a cryptographic guarantee, and it inherits the baseline's open access-pattern channel. Closing the residual graph and the overlap channel with the micro-key and coarse-cell-ID variants, stress-testing Shard against learned and zero-shot inverters, and composing it with a PIR-style access-pattern hider are the natural next steps. The contribution here is to move embedding privacy from a single alignable geometry, optimised against a distortion surrogate, to a *family* of cell-keyed transforms whose advantage against the modern attack surface is measured rather than assumed.

Reproducibility

All scripts, notebooks, configuration files and JSON/CSV outputs of the experiments are packaged as supplementary material for review. The revision adds `notebooks/exp7_alignment_pq_leakage.py` for the known-plaintext and PQ-leakage experiments, `notebooks/exp8_tradeoff_noise_sweep.py` for the SVD/noise utility–leakage diagnostic, `notebooks/exp9_reference_` corpus_attack.py for the reference-corpus lookup stress test, notebooks/exp10_denoiser_significance.py for the paired significance tests of Section 8.5 (McNemar and bootstrap, computed from the cached embeddings with numpy/scipy only), notebooks/exp11_beir_denoiser.py for the BEIR generality experiment of Section 8.6 (CPU-only encoding of SciFact/NFCorpus), notebooks/shard_lib.py for the SHARD construction and notebooks/exp12_shard_utility.py, notebooks/exp13_shard_alignment.py, notebooks/exp14_shard_leakage.py, notebooks/exp15_shard_reference.py, notebooks/exp17_beir_shard.py, notebooks/exp18_shard_cost.py (active-cells/bandwidth), notebooks/exp19_shard_targeted.py (targeted attacker) and notebooks/exp20_shard_microkey.py (micro-key leak/unlinkability), notebooks/exp21_shard_vs_dp.py (vs. distortion-aware DP-noise) and notebooks/exp22_shard_learned_attack.py (learned/unsupervised attackers) for the SHARD utility, alignment anchor-complexity (diffuse, targeted and learned), public-index leakage, reference-lookup, BEIR, online-cost, micro-key and defence-comparison experiments (numpy/scikit-learn only except the CPU BEIR encoding; deterministic master-key seeds; e5-small and e5-base), with figures from notebooks/make_fig_shard.py, notebooks/make_paper_figures.py for regenerating every figure in this paper, and notebooks/heavy_adaptive_inversion_rtx4090.py plus notebooks/requirements_rtx4090.txt for the aligned Vec2Text stress test sized for an RTX 4090 with about 11 GiB of free VRAM. The resulting case checkpoints and final summaries are stored in exp8_adaptive_outputs_rtx4090/, notebooks/exp10_outputs/ and notebooks/exp11_outputs/. The integral and significance experiments are driven by deterministic seeds (query seed 42, rotation seeds $\{11, 23, 31, 47, 53\}$, bootstrap seed 2026) and JSON output files; the parameter-selection grid ships as CSV. The corpus (one-million Russian-Wikipedia paragraphs) is reproducibly sliced from a public Wikipedia dump with the date and content checksum recorded in the data manifest; the BEIR data is the public BeIR/scifact and BeIR/nfcorpus releases. A public DOI archive and target-journal data statement remain to be added before formal submission.

References

[1] N. Reimers and I. Gurevych, "Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks," in *Proc. EMNLP*, 2019, pp. 3982–3992.

[2] L. Wang, N. Yang, X. Huang, et al., "Text Embeddings by Weakly-Supervised Contrastive Pre-training," *arXiv:2212.03533*, 2022.

[3] J. Ni et al., "Large Dual Encoders Are Generalizable Retrievers," in *Proc. EMNLP*, 2022, pp. 9844–9855.

[4] J. Chen, S. Xiao, P. Zhang, K. Luo, D. Lian, and Z. Liu, "BGE M3-Embedding: Multi-Lingual, Multi-Functionality, Multi-Granularity Text Embeddings Through Self-Knowledge Distillation," *arXiv:2402.03216*, 2024.

[5] V. Karpukhin et al., "Dense Passage Retrieval for Open-Domain Question Answering," in *Proc. EMNLP*, 2020, pp. 6769–6781.

[6] O. Khattab and M. Zaharia, "ColBERT: Efficient and Effective Passage Search via Contextualised Late Interaction over BERT," in *Proc. ACM SIGIR*, 2020, pp. 39–48.

[7] J. X. Morris, V. Kuleshov, V. Shmatikov, and A. M. Rush, "Text Embeddings Reveal (Almost) As Much As Text," in *Proc. EMNLP*, 2023.

[8] J. Gu et al., "GEIA: Gradient-based Embedding Inversion Attack," *arXiv:2304.08945*, 2023.

[9] X. Pan et al., "TEIA: Text Embedding Inversion Attacks," *arXiv:2305.08450*, 2023.

[10] Y. Chen, Q. Xu, and J. Bjerva, "ALGEN: Few-shot Inversion Attacks on Textual Embeddings via Cross-Model Alignment and Generation," in *Proc. ACL*, 2025.

[11] Y. Zhang, J. X. Morris, and V. Shmatikov, "Universal Zero-shot Embedding Inversion," *arXiv:2504.00147*, 2025.

[12] Y. Yu et al., "LAGO: Few-shot Crosslingual Embedding Inversion Attacks via Language Similarity-Aware Graph Optimization," *arXiv:2505.16008*, 2025.

[13] T. Liu et al., "EGuard: Defending LLM Embeddings Against Inversion Attacks via Text Mutual Information Optimization," in *Proc. AAAI*, 2026.

[14] Y.-C. Tsai, H. Hsiao, K.-Y. Chen, and S.-D. Lin, "Concept-Aware Privacy Mechanisms for Defending Embedding Inversion Attacks," in *Proc. ICLR*, 2026.

[15] D. El Zein, S. Kumar, and J. Henderson, "Nonparametric Variational Differential Privacy via Embedding Parameter Clipping," *arXiv:2603.09583*, 2026.

[16] A. Kusupati et al., "Matryoshka Representation Learning," in *Advances in Neural Information Processing Systems*, 2022.

[17] R. Jha, C. Zhang, V. Shmatikov, and J. X. Morris, "Harnessing the Universal Geometry of Embeddings," *arXiv:2505.12540*, 2025.

[18] S. Zeng et al., "The Good and the Bad: Exploring Privacy Issues in Retrieval-Augmented Generation (RAG)," in *Findings of ACL*, 2024, pp. 4505–4524.

[19] C. Song and A. Raghunathan, "Information Leakage in Embedding Models," in *Proc. ACM CCS*, 2020, pp. 377–390.

[20] R. Shokri et al., "Membership Inference Attacks Against Machine Learning Models," in *Proc. IEEE S&P*, 2017, pp. 3–18.

[21] N. Carlini et al., "Extracting Training Data from Large Language Models," in *Proc. USENIX Security*, 2021, pp. 2633–2650.

[22] C. Dwork, "Differential Privacy," in *Proc. ICALP*, 2006, pp. 1–12.

[23] M. Abadi et al., "Deep Learning with Differential Privacy," in *Proc. ACM CCS*, 2016, pp. 308–318.

[24] L. Lyu, X. He, and Y. Li, "Differentially Private Representation for NLP," in *Findings of EMNLP*, 2020, pp. 2355–2365.

[25] K. Kenthapadi, A. Korolova, I. Mironov, and N. Mishra, "Privacy via the Johnson-Lindenstrauss Transform," *J. Privacy and Confidentiality*, vol. 5, no. 1, 2013.

[26] K. Liu, H. Kargupta, and J. Ryan, "Random Projection-Based Multiplicative Data Perturbation for Privacy Preserving Distributed Data Mining," *IEEE TKDE*, vol. 18, no. 1, 2006, pp. 92–106.

[27] J. H. Cheon, A. Kim, M. Kim, and Y. Song, "Homomorphic Encryption for Arithmetic of Approximate Numbers," in *Advances in Cryptology—ASIACRYPT 2017*, LNCS 10624, pp. 409–437.

[28] J. H. Cheon, K. Han, A. Kim et al., "Bootstrapping for Approximate Homomorphic Encryption," in *Advances in Cryptology—EUROCRYPT 2018*, LNCS 10820, pp. 360–384.

[29] M. Albrecht et al., "Homomorphic Encryption Security Standard," HomomorphicEncryption.org, 2018.

[30] M. R. Albrecht, R. Player, and S. Scott, "On the Concrete Hardness of Learning with Errors," *Journal of Mathematical Cryptology*, vol. 9, no. 3, 2015, pp. 169–203 (lattice-estimator methodology).

[31] P. H. Schönemann, "A Generalized Solution of the Orthogonal Procrustes Problem," *Psychometrika*, vol. 31, no. 1, 1966, pp. 1–10.

[32] V. M. Patel, N. K. Ratha, and R. Chellappa, "Cancelable Biometrics: A Review," *IEEE Signal Processing Magazine*, vol. 32, no. 5, 2015, pp. 54–65.

[33] A. Al Badawi et al., "OpenFHE: Open-Source Fully Homomorphic Encryption Library," in *Proc. WAHC '22*, 2022, pp. 53–63.

[34] R. Dathathri et al., "EVA: An Encrypted Vector Arithmetic Language and Compiler for Efficient Homomorphic Computation," in *Proc. ACM PLDI*, 2020, pp. 546–561.

[35] R. Dathathri et al., "CHET: An Optimizing Compiler for Fully-Homomorphic Neural-Network Inferencing," in *Proc. ACM PLDI*, 2019, pp. 142–156.

[36] W. Jung, S. Kim, J. H. Ahn et al., "Over 100x Faster Bootstrapping in Fully Homomorphic Encryption through Memory-centric Optimisation with GPUs," *IACR TCHES*, vol. 2021, no. 4, pp. 114–148.

[37] F. Boemer et al., "Intel HEXL: Accelerating Homomorphic Encryption with Intel AVX512-IFMA52," in *Proc. WAHC '21*, 2021, pp. 57–62.

[38] A. Henzinger, E. Dauterman, H. Corrigan-Gibbs, and N. Zeldovich, "Private Web Search with Tiptoe," in *Proc. ACM SOSP*, 2023.

[39] S. Angel, H. Chen, K. Laine, and S. Setty, "PIR with Compressed Queries and Amortised Query Processing," in *Proc. IEEE S&P*, 2018, pp. 962–979.

[40] A. Henzinger, M. M. Hong, H. Corrigan-Gibbs, S. Meiklejohn, and V. Vaikuntanathan, "One Server for the Price of Two: Simple and Fast Single-Server Private Information Retrieval (SimplePIR)," in *Proc. USENIX Security*, 2023.

[41] M. H. Mughees, H. Chen, and L. Ren, "OnionPIR: Response Efficient Single-Server PIR," in *Proc. ACM CCS*, 2021, pp. 2292–2306.

[42] N. Halko, P.-G. Martinsson, and J. A. Tropp, "Finding Structure with Randomness: Probabilistic Algorithms for Constructing Approximate Matrix Decompositions," *SIAM Review*, vol. 53, no. 2, 2011, pp. 217–288.

[43] H. Jegou, M. Douze, and C. Schmid, "Product Quantisation for Nearest Neighbor Search," *IEEE TPAMI*, vol. 33, no. 1, 2011, pp. 117–128.

[44] C. Eckart and G. Young, "The Approximation of One Matrix by Another of Lower Rank," *Psychometrika*, vol. 1, no. 3, 1936, pp. 211–218.

[45] W. B. Johnson and J. Lindenstrauss, "Extensions of Lipschitz Mappings into a Hilbert Space," *Contemporary Mathematics*, vol. 26, 1984, pp. 189–206.

[46] J. H. Friedman, "Greedy Function Approximation: A Gradient Boosting Machine," *Annals of Statistics*, vol. 29, no. 5, 2001, pp. 1189–1232.

[47] K. Papineni et al., "BLEU: a Method for Automatic Evaluation of Machine Translation," in *Proc. ACL*, 2002, pp. 311–318.

[48] P. Lewis et al., "Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks," in *Proc. NeurIPS*, 2020.

[49] N. Thakur, N. Reimers, A. Rücklé, A. Srivastava, and I. Gurevych, "BEIR: A Heterogeneous Benchmark for Zero-shot Evaluation of Information Retrieval Models," in *Proc. NeurIPS Datasets and Benchmarks*, 2021.